Materials Science 1 – Dislocation motion and Strengthening Mechanisms

Learning objectives and Learning outcomes

- Describe edge and screw dislocation motion from an atomic perspective.
- Describe how plastic deformation occurs by the motion of edge and screw dislocations in response to applied shear stresses.
- Sketch the process of deformation of single crystals.
- Determine the critical resolved shear stress of materials.
- List the processes that can be used to strengthen materials.

Poll – Answers in We Chat

Q1) Edge dislocations and screw dislocation are linear crystalline defects.

A) True;

B) False.

Poll – Answer

Q1) Edge dislocations and screw dislocation are linear crystalline defects.

A) True.

Edge dislocations and screw dislocation are line-defects in crystalline solids.

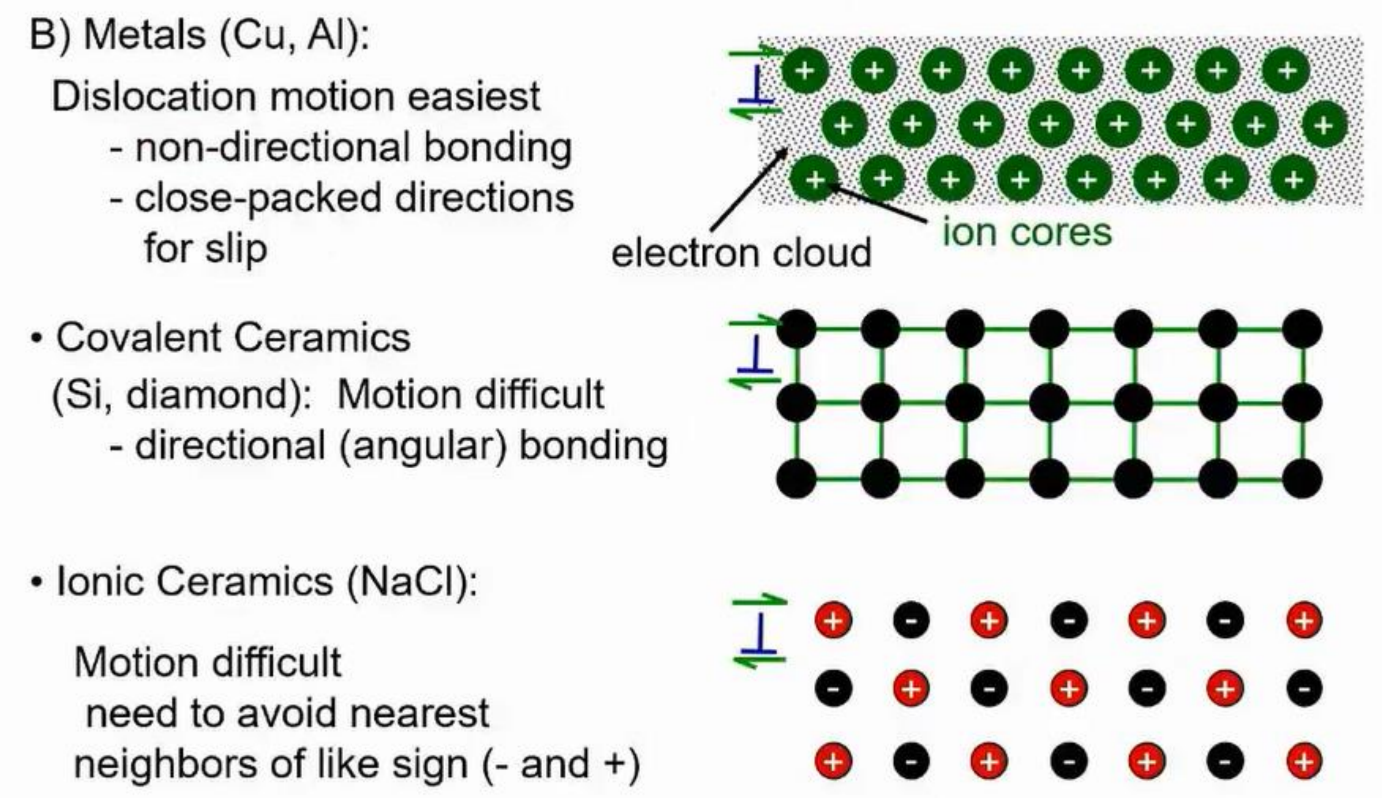
Poll – Answers in We Chat

Q1) For which family of materials will dislocation motion be favoured?

A) Polymers
B) Metals
C) Ceramics

Poll – Answer

Q1) For which family of materials will dislocation motion be favoured?



- Polymer chains are long and flexible; they can bend, twist and slide past one another rather easily, so they can accommodate deformation without the need for dislocation motion. The movement of polymer chains does not follow well-defined slip systems.

Origin of dislocations in metals

Dislocations are introduced by various processes:

- During solidification;
- Induced by plastic deformation;
- Induced as a consequence of thermal stresses that result from rapid cooling.

Dislocation Density

The dislocation density in a material is expressed as the total dislocation length per unit volume or, equivalently, as the number of dislocations that intersect a unit area of a random section. The units of dislocation density are [mm/mm^3], or just [1/mm^2].

- Low dislocation densities in the order of 100 mm^{-2} are typically found in carefully solidified metal crystals.

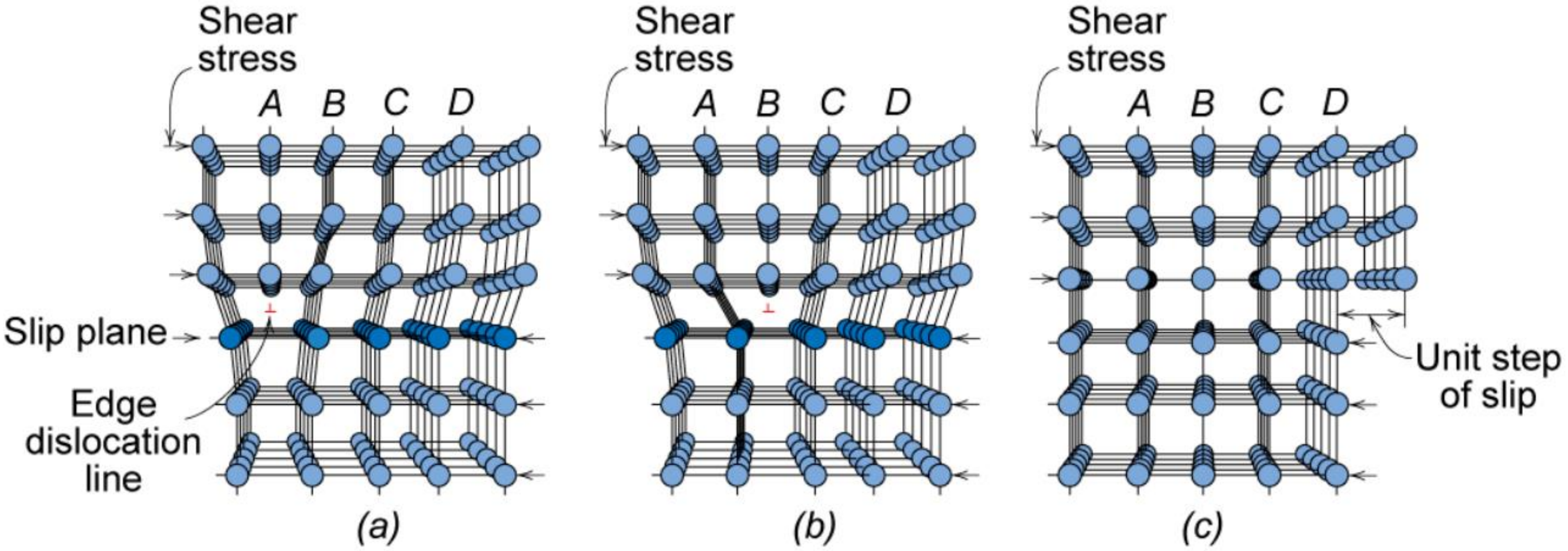
- For heavily deformed metals using various procedure (i.e. cold roll, cold forging, etc) dislocation density can achieve 1000 mm^{-2} .

- Specific heat-treatments at temperature higher than the so called “recrystallization temperature” can significantly reduce dislocation density.

Dislocation motion – atomic level

In metals, plastic deformation occurs by the motion of large numbers of dislocations.

An edge dislocation moves in response to a shear stress applied perpendicularly to its line.

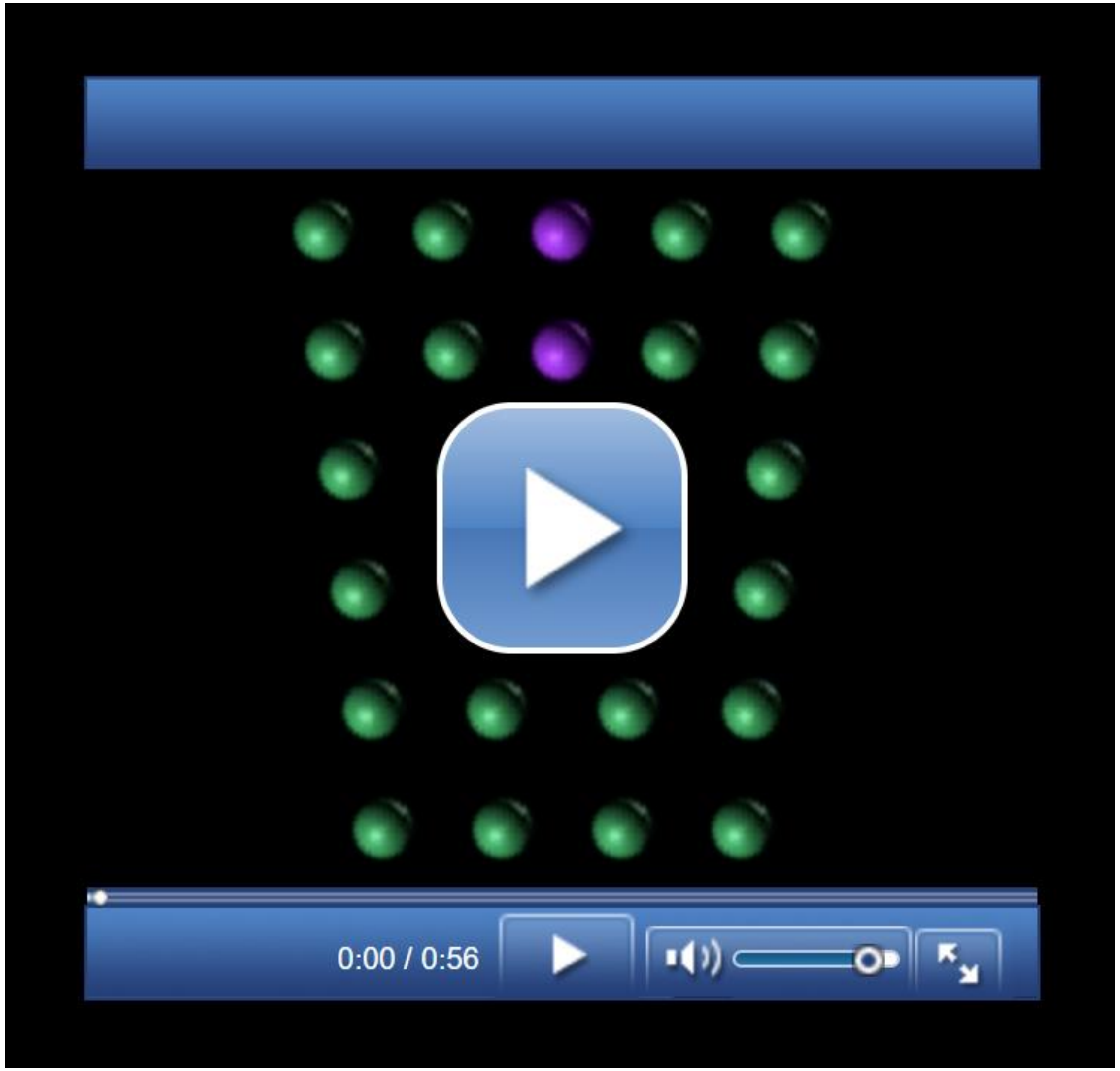


When the shear stress is applied, the plane A is forced to the right, pushing the top halves of planes B, C, D in the same direction. If the shear stress is sufficient, the plane A links up with the initial bottom half of plane B.

Fig. 7.1, Callister & Rethwisch 9e.

- This **"slip" process** is repeated for the other planes by successive and repeated breaking of bonds and shifting by interatomic distances. The slip direction is the same as the Burgers vector direction.
- By moving on the **slip plane**, the extra half-plane may emerge from the right surface of the crystal, forming an edge that is one atomic distance wide.

Dislocation motion – slip process animation

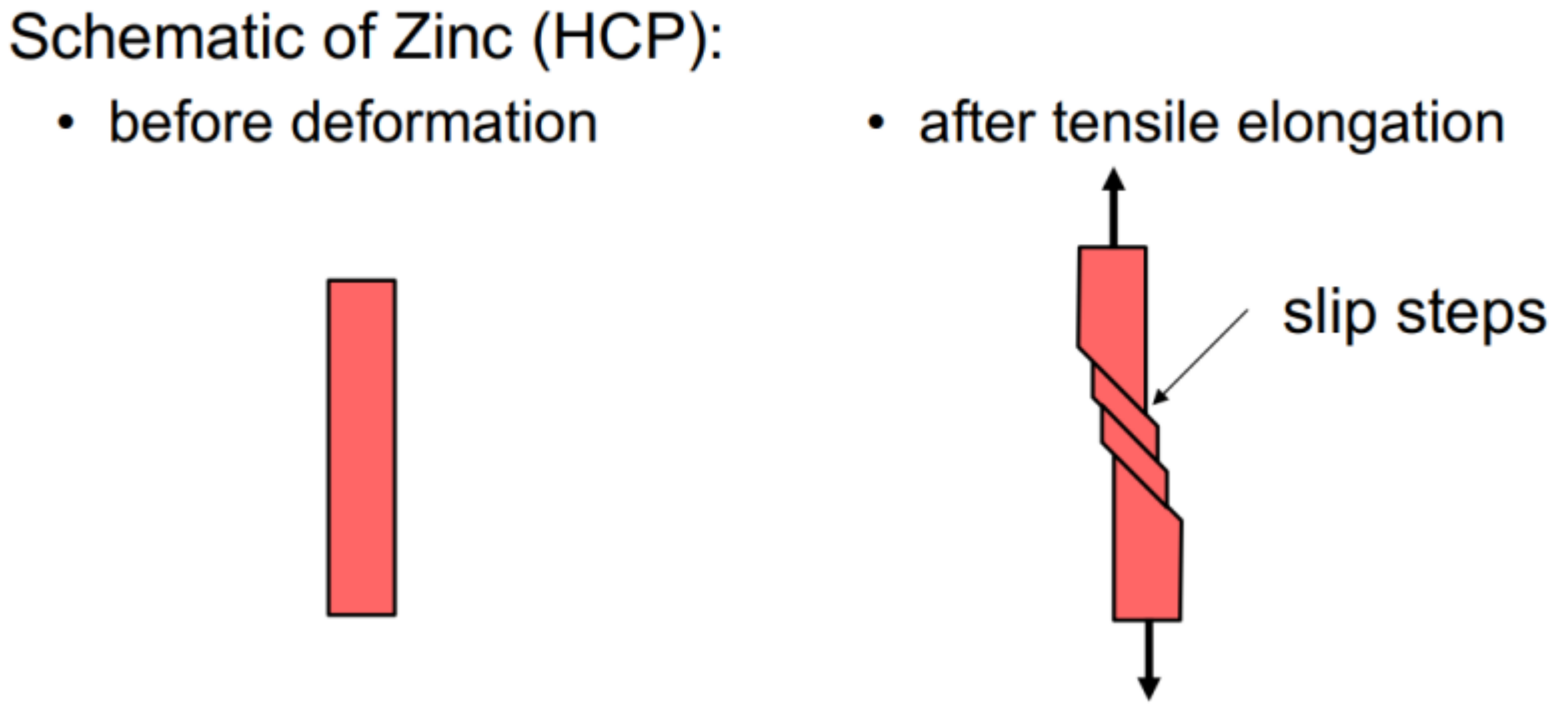


Wiley - VMSE (drbuc2jl8158i.cloudfront.net)

Other types of imperfections in solids

Movement of dislocations

Macroscopically, the movement of dislocation on the slip planes produces plastic deformations (permanent or irreversible deformations).



Dislocation motion – slip systems

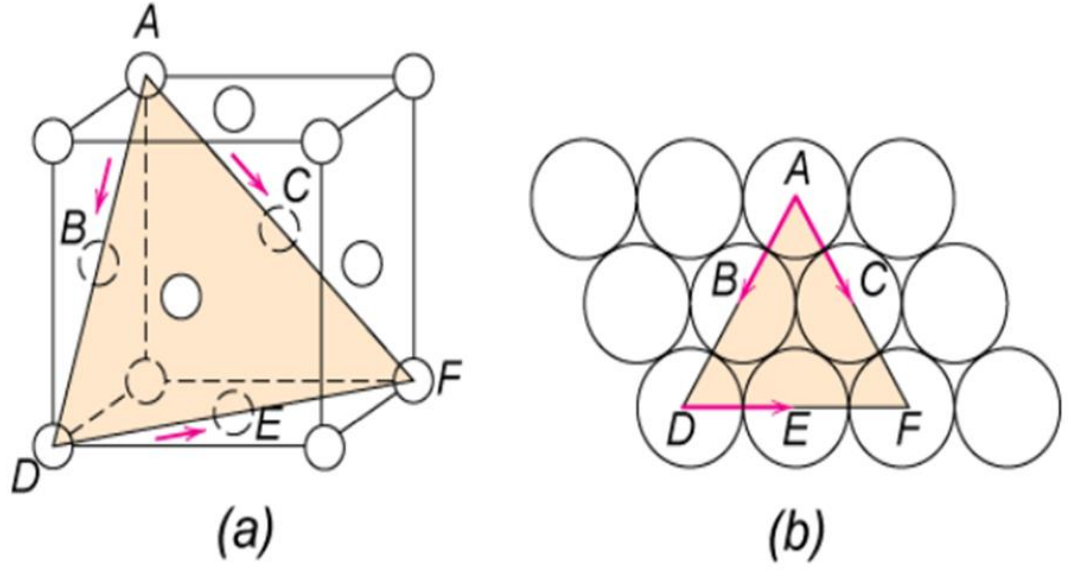
Dislocations motion is favoured on specific crystallographic planes, called **slip planes**, and along specific crystal directions, known as **slip directions**.

Slip System

- Slip plane - plane on which easiest slippage occurs
  - Highest planar densities (and large interplanar spacings)
- Slip directions - directions of movement
  - Highest linear densities

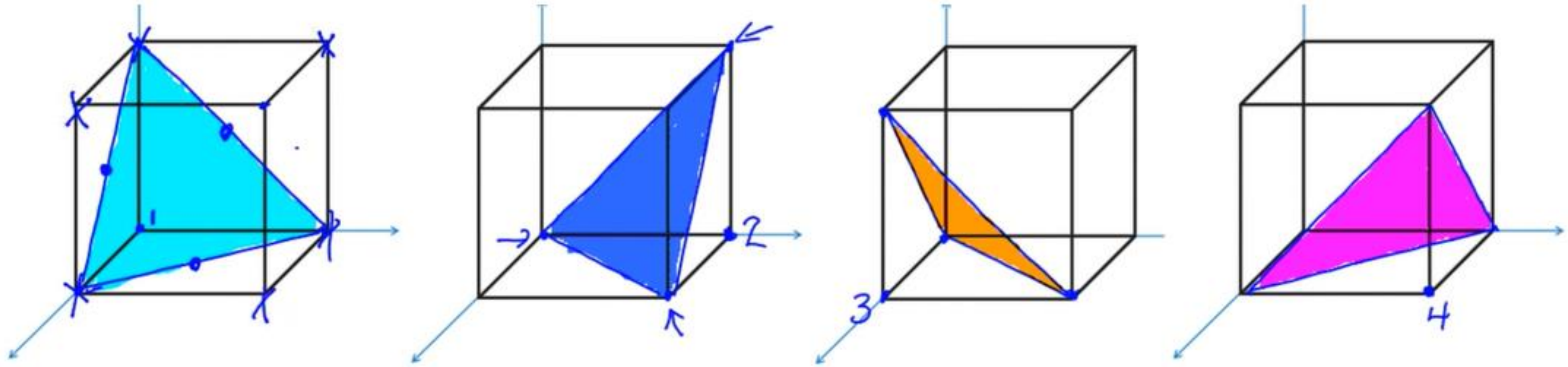
Dislocation motion – slip systems

Slip systems in the FCC structure.



(a) A {111} ⟨110⟩ slip system shown within an FCC unit cell. (b) The (111) plane from (a) and three ⟨110⟩ slip directions (as indicated by arrows) within that plane constitute possible slip systems.

There are 3 [110] directions on each of the 4 non-parallel (111) planes. These 4 planes can be identified by moving the origin of the axes in the points 1, 2, 3 and 4.



The total number of slip systems in FCC structure is 3x4=12.

Dislocation motion – slip systems

Slip systems in various structures.

Metals	*Slip Plane*	*Slip Direction*	*Number of Slip Systems*
	Face-Centered Cubic		
Cu, Al, Ni, Ag, Au	{111}	⟨110⟩	12
	Body-Centered Cubic		
α-Fe, W, Mo	{110}	⟨111⟩	12
α-Fe,W	{211}	⟨111⟩	12
α-Fe, K	{321}	⟨111⟩	24
	Hexagonal Close-Packed		
Cd, Zn, Mg, Ti, Be	{0001}	$\langle 11\bar{2}0\rangle$	3
Ti, Mg, Zr	$\{10\bar{1}0\}$	$\langle 11\bar{2}0\rangle$	3
Ti, Mg	$\{10\bar{1}1\}$	$\langle 11\bar{2}0\rangle$	6

Poll – Answers in We Chat

Q1) Why metals with HCP crystal structure are more brittle than metals with FCC or BCC structures?

A) Because they have covalent bonding;
B) Because they have more slip systems;
C) Because they have less slip systems;
D) Because they have ionic bonding.

Poll – Answer

Q1) Why metals with HCP crystal structure are more brittle than metals with FCC or BCC structures?

C) There are less possible planes and directions on which dislocation motion occurs in HCP crystals, resulting in lower plastic deformation and therefore materials with HCP structure are more brittle.

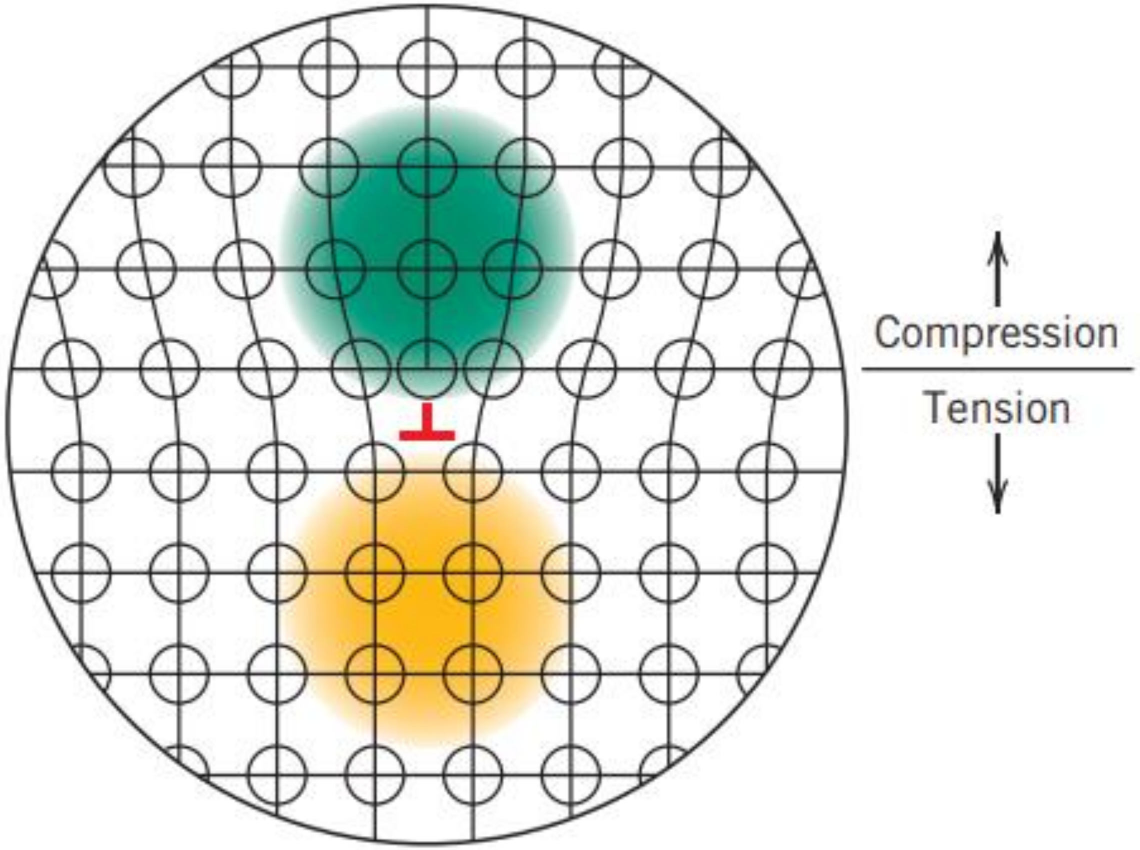
Metals	*Slip Plane*	*Slip Direction*	*Number of Slip Systems*
	Face-Centered Cubic		
Cu, Al, Ni, Ag, Au	{111}	$\langle 1\bar{1}0 \rangle$	12
	Body-Centered Cubic		
α-Fe, W, Mo	{110}	$\langle \bar{1}11 \rangle$	12
α-Fe, W	{211}	$\langle \bar{1}11 \rangle$	12
α-Fe, K	{321}	$\langle \bar{1}11 \rangle$	24
	Hexagonal Close-Packed		
Cd, Zn, Mg, Ti, Be	{0001}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg, Zr	$\{10\bar{1}0\}$	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg	$\{10\bar{1}1\}$	$\langle 11\bar{2}0 \rangle$	6

Table 9.1 Callister & Rethwish, 9e

Stress and strain fields around a dislocation

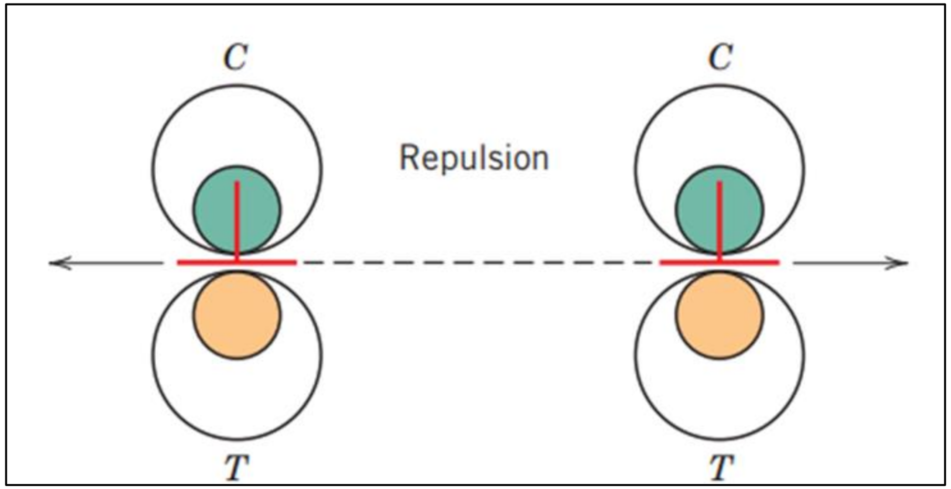
The presence of a dislocation induces an atomic lattice distortion, which causes an alteration of the strain and stress profile in the lattice.

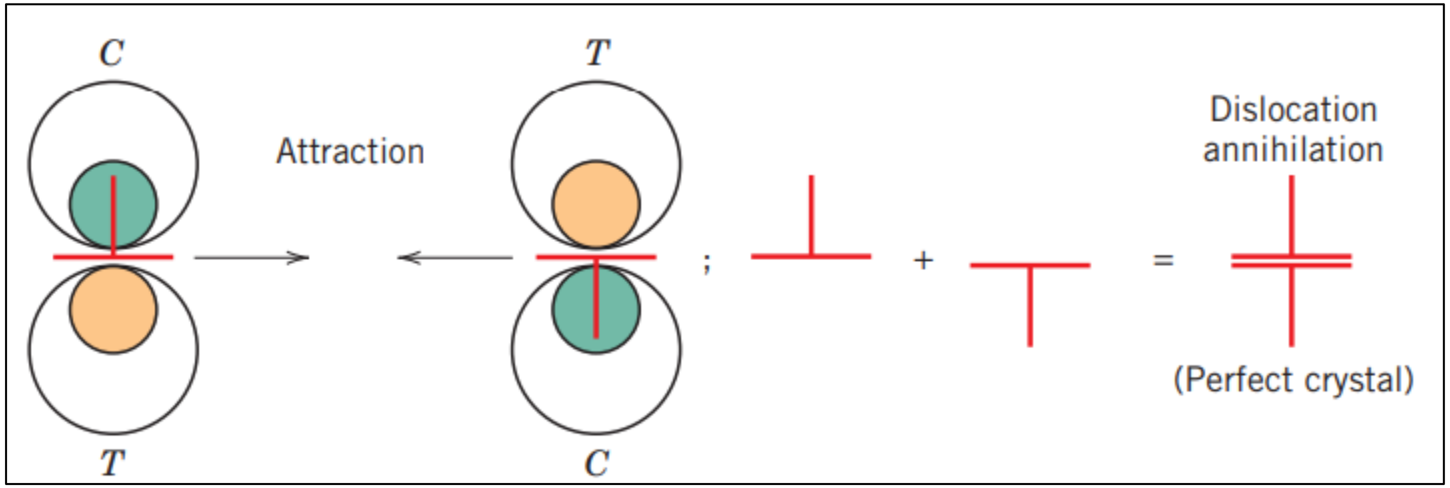
- For an edge dislocation, atoms above and adjacent to the dislocation line are squeezed, and experience a **compressive strain** compared to the perfect crystal.
- Directly below the half-plane, the lattice atoms experience **tensile strain**.
- Shear strains also exist in the vicinity of the edge dislocation.
- For a screw dislocation, lattice strains are pure shear only.



Interactions between dislocations

Dislocations interact within the lattice; their interaction also depends on their orientations.





Two edge dislocations of the same sign and lying on the same slip plane exert a repulsive force on each other.

Edge dislocations of opposite sign and lying on the same slip plane exert an attractive force on each other. Upon meeting, they annihilate each other and leave a region of perfect crystal.

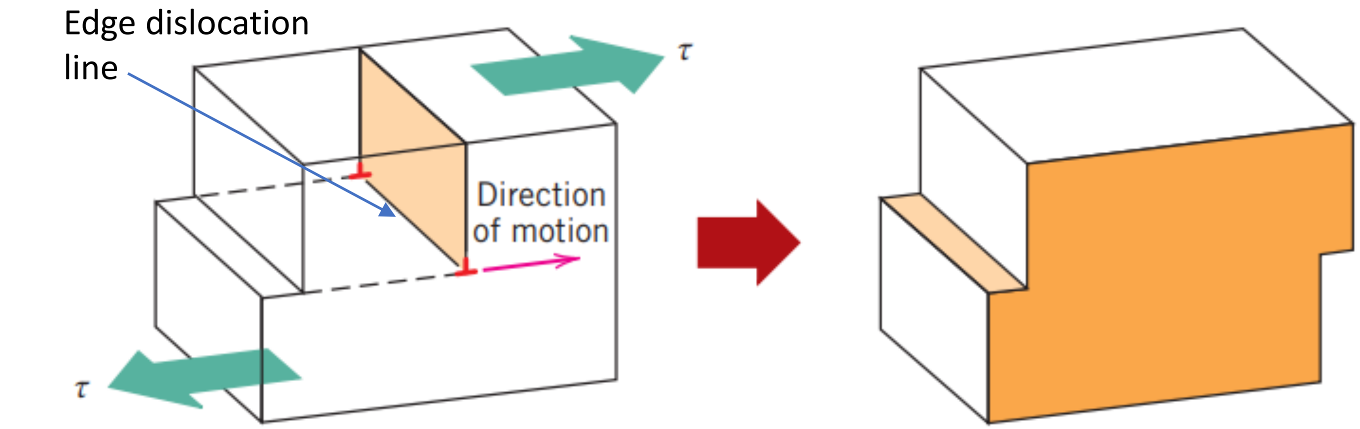
Macroscopic effects of dislocation motion

- For an edge, the dislocation line moves in the direction of the applied shear stress τ (Fig.a).

- For a screw, the dislocation line motion is perpendicular to the stress direction (Fig.b).

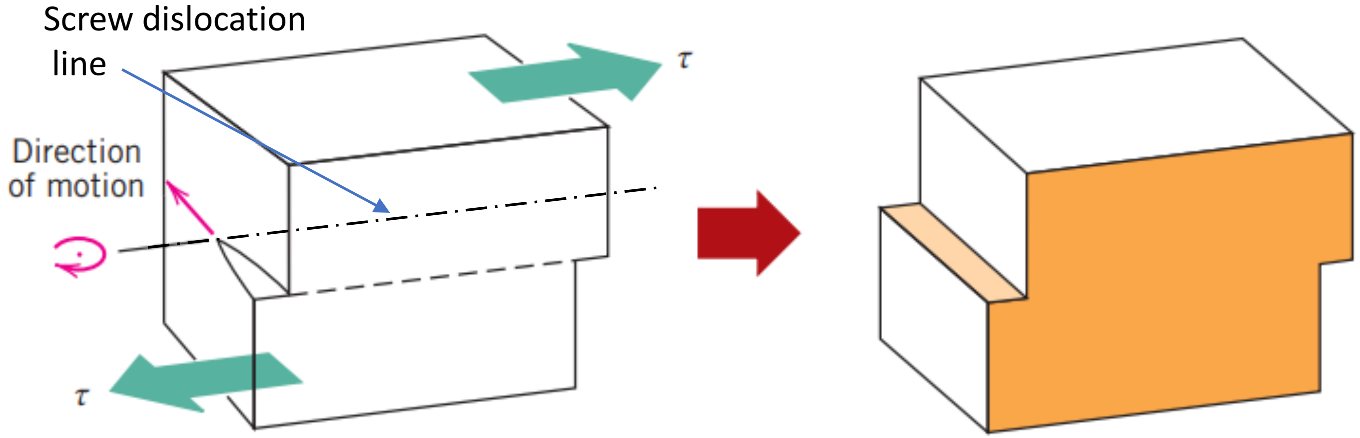
- The net plastic deformation for the motion of both dislocation types is the same.

Edge dislocation



(a)

Screw dislocation



(b)

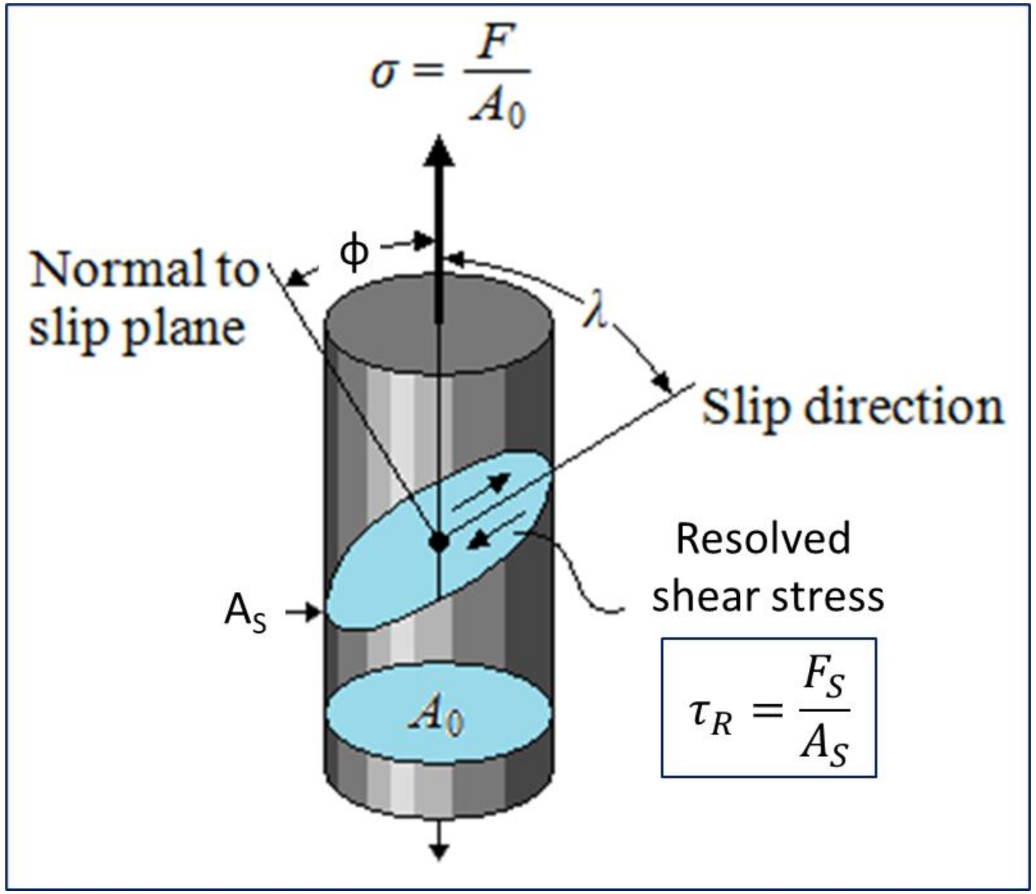
Fig. 7.2, Callister & Rethwisch 9e.

Slip in single crystals – Resolved shear stress

The description of slip process in materials is simplified in single crystals.

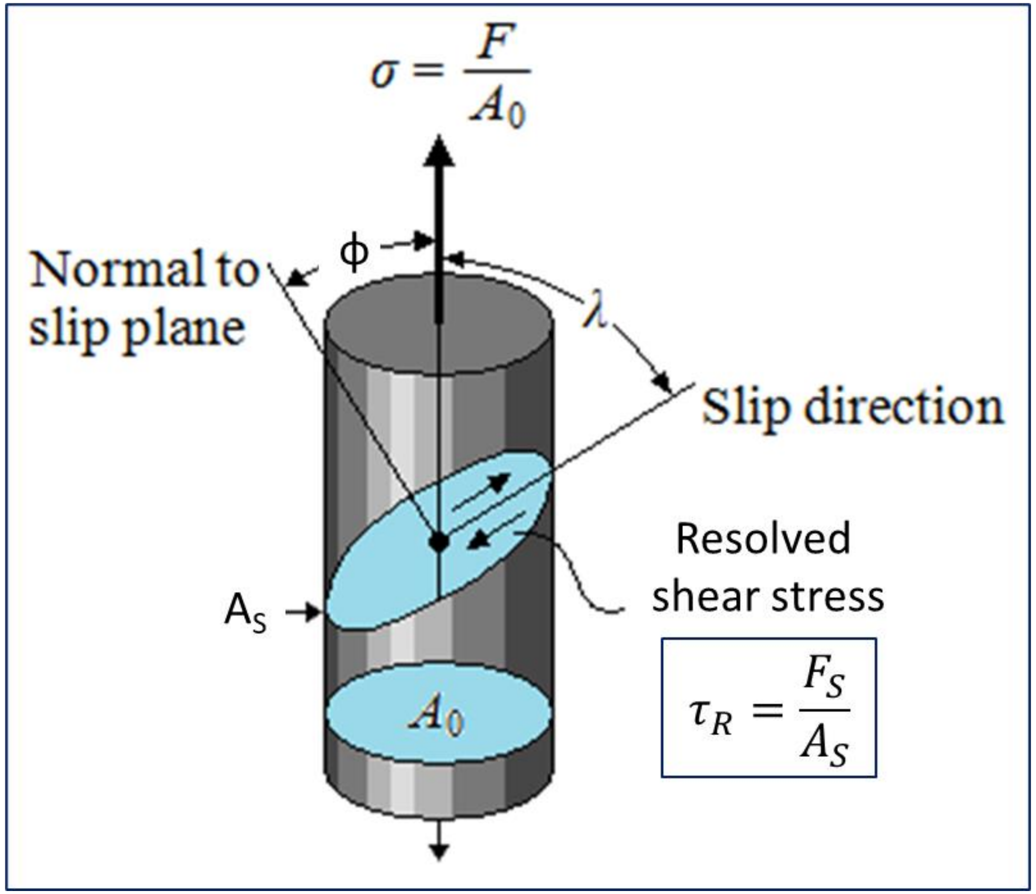
Imagine a tensile or compressive load applied to single crystal specimen. The applied load will have force components that would exert shear stresses on crystallographic slip planes, which can be oriented with a specific angle along the loading direction.

These shear stress are known as **resolved shear stresses**, and their magnitudes depend on the applied stress and the orientation of both the slip plane and direction within that plane.



Slip in single crystals – Resolved shear stress

If ϕ represents the angle between the normal to the slip plane and the applied stress direction, and λ the angle between the slip direction and the applied stress direction.



$$\sigma = \frac{F}{A_0}$$

$$F_S = \mathrm{F}\cos\lambda$$

$$A_S = \frac{A_0}{\cos\phi}$$

Resolved shear stresses

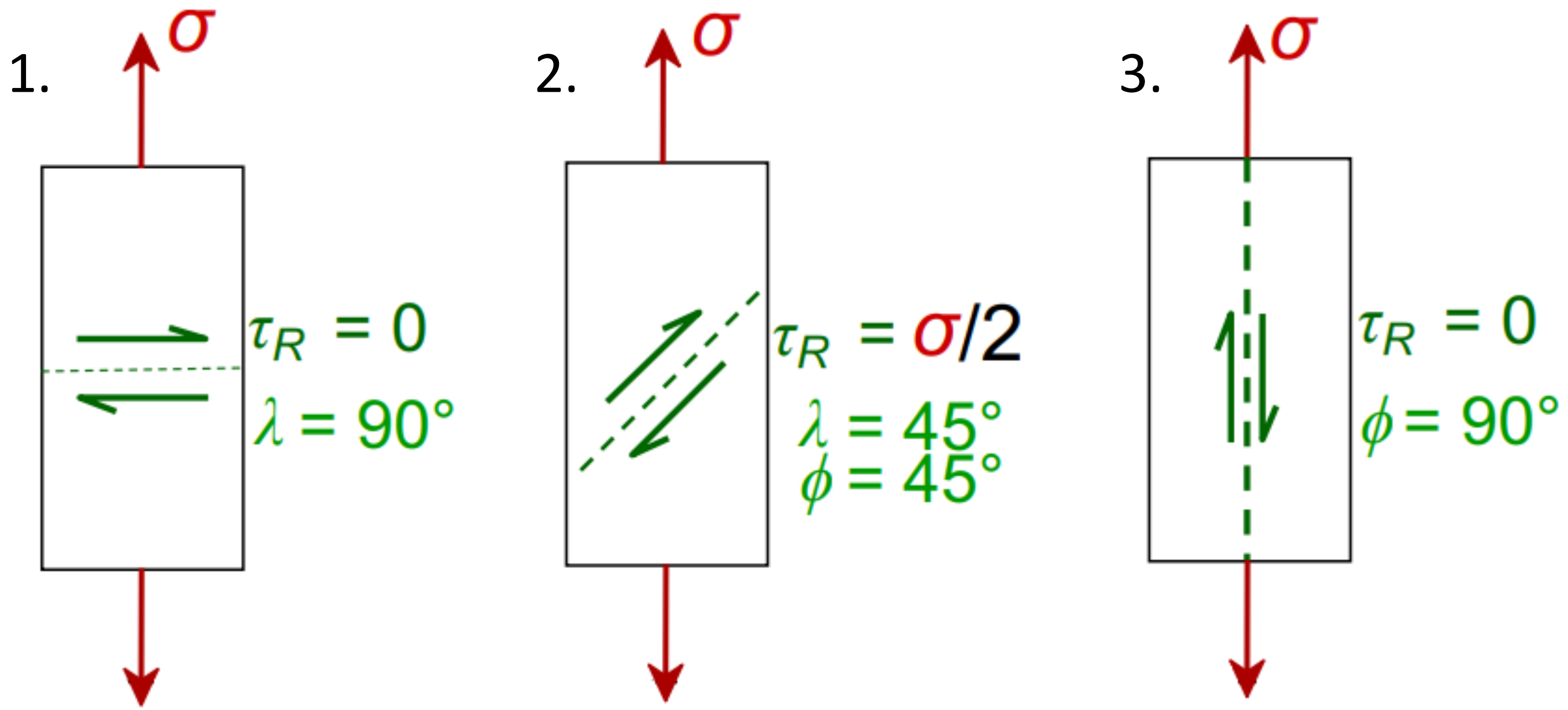
$$\boxed{\tau_R = \frac{F_S}{A_S} = \sigma\cos\lambda\cos\phi}$$

Slip in single crystals – Resolved shear stress

According to the relationship between the applied normal stress σ and the resolved shear stress τ_R:

$$\boxed{\tau_R = \sigma cos\lambda cos\phi}$$

Three limit cases can be considered:



Maximum τ_R for
$\lambda = \phi = 45°$

Slip in single crystals – Critical resolved shear stress τ_{CRSS}

The slip process and consequent plastic deformation begin on the most favourably oriented slip system when the resolved shear stress on that specific plane will reach a critical value known as **critical resolved shear stress τ_{crss}**.

This parameters represents the minimum shear stress required to initiate slip and is a property of the material that determines when yielding occurs.

Since: $\boxed{\tau_R(max) = \sigma(cos\lambda\cos\phi)_{max} = \frac{\sigma}{2}}$ Which means that τ_R is maximum for $\lambda = \phi$=45°.

Since plastic deformation occur for $\sigma = \sigma_Y$, it follows that: $\boxed{\tau_{CRSS} = \frac{\sigma_Y}{2}}$

Slip in single crystals – Macroscopic effects

- Slip occurs along a number of equivalent and most favourably oriented planes and directions at various positions along the specimen length.

- The slip process will cause slip lines to appear on the specimen's surface.

- With increasing deformation, the number of slip lines and slip step width will increase.

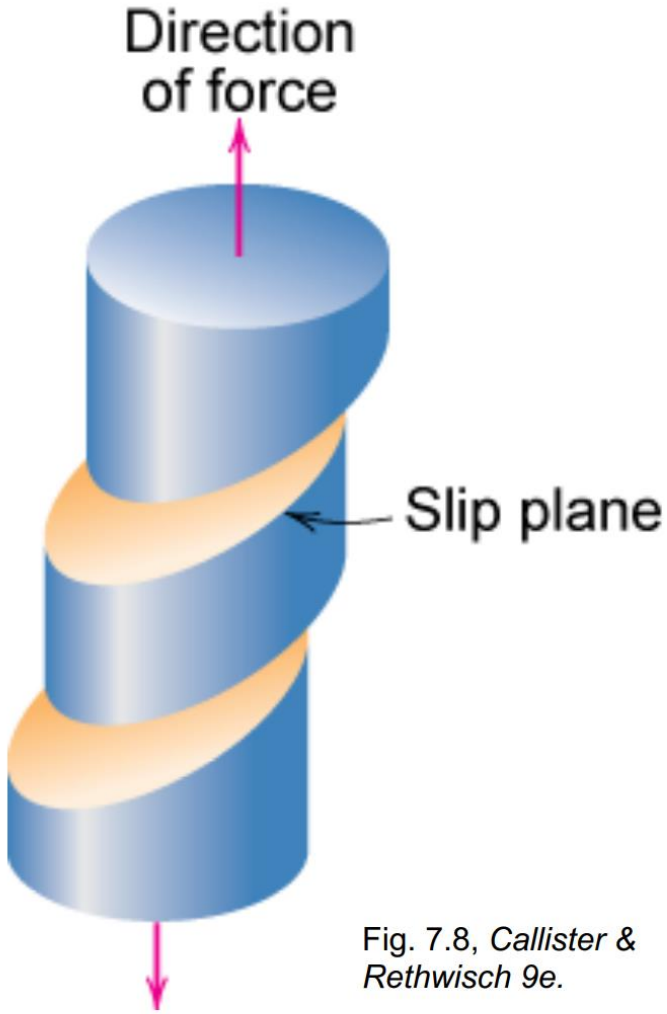
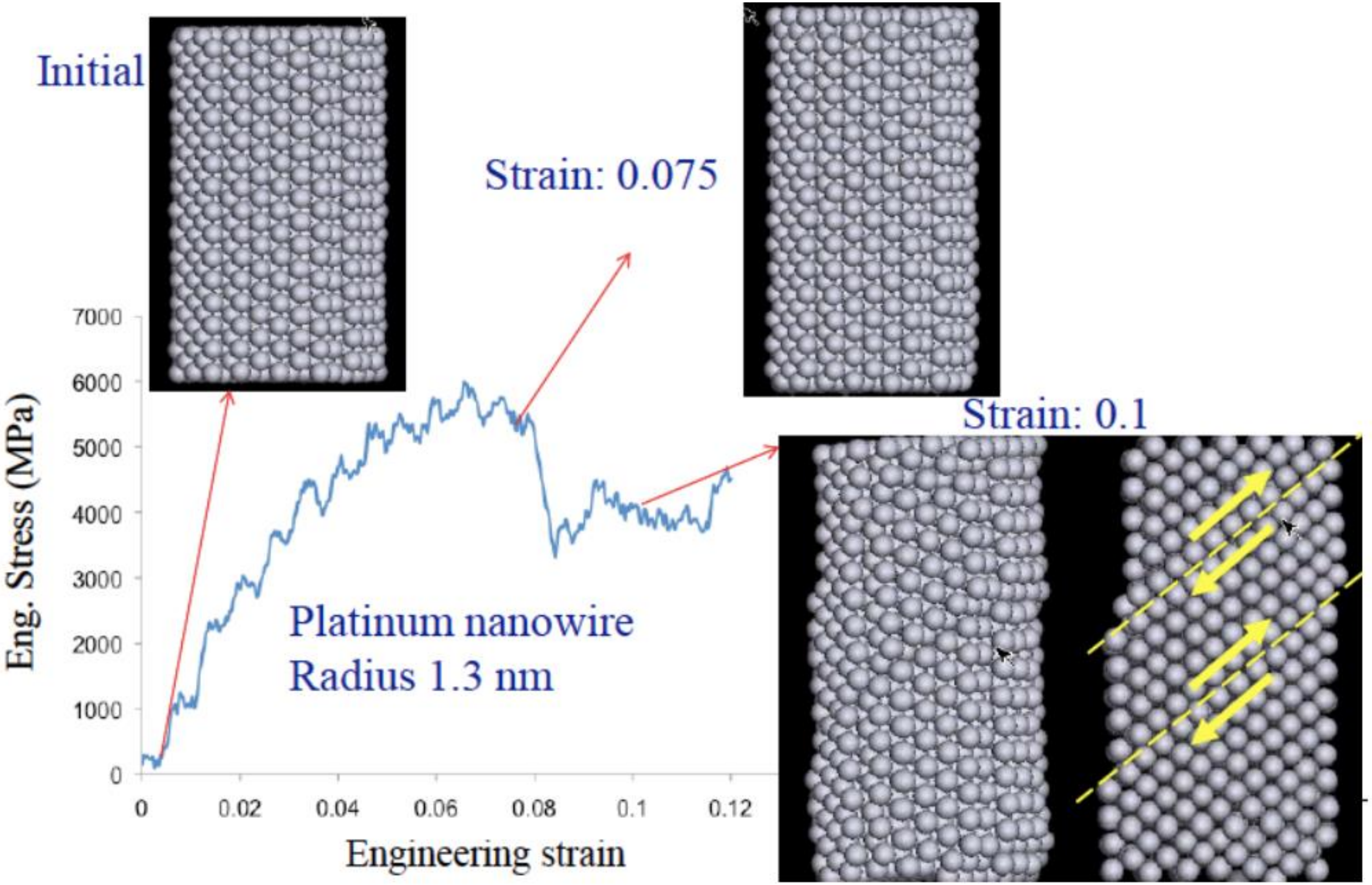


Fig. 7.8, *Callister & Rethwisch 9e.*

Fig. 7.9, *Callister & Rethwisch 9e.* (From C. F. Elam, *The Distortion of Metal Crystals*, Oxford University Press, London, 1935.)

Zinc single crystal that has been plastically deformed.

Atomic simulations of slip process in platinum nanowire under tensile stress



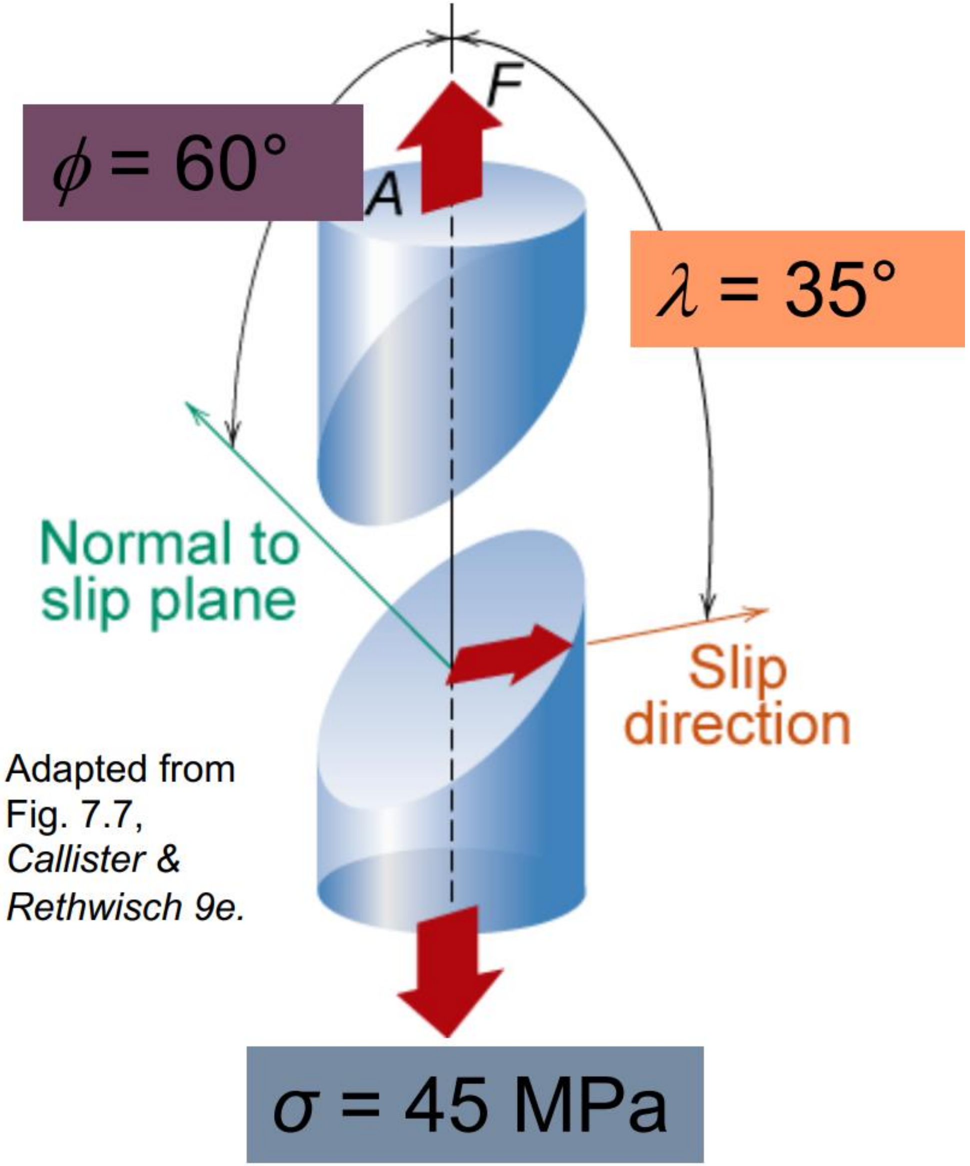
Slip in single crystals

Example 1:

Consider a metal single crystal specimen subjected to a normal stress of $\sigma = 45\ \mathrm{MPa}$. The normal to the slip plane forms an angle ϕ=60° with the loading direction and the slip direction forms an angle of λ=35° with the loading direction. If the critical resolved shear stress is $\tau_{CRSS} = 20.7\ \mathrm{Mpa}$, determine:

a) if the single crystal will yield (experience plastic deformations);

b) If it does not yield under the present loading conditions, determined the stress needed to induce yield.



Adapted from Fig. 7.7, *Callister & Rethwisch 9e.*

Slip in single crystals

Example 1 solution:

a) The resolved shear stress can be calculated using the relationship:

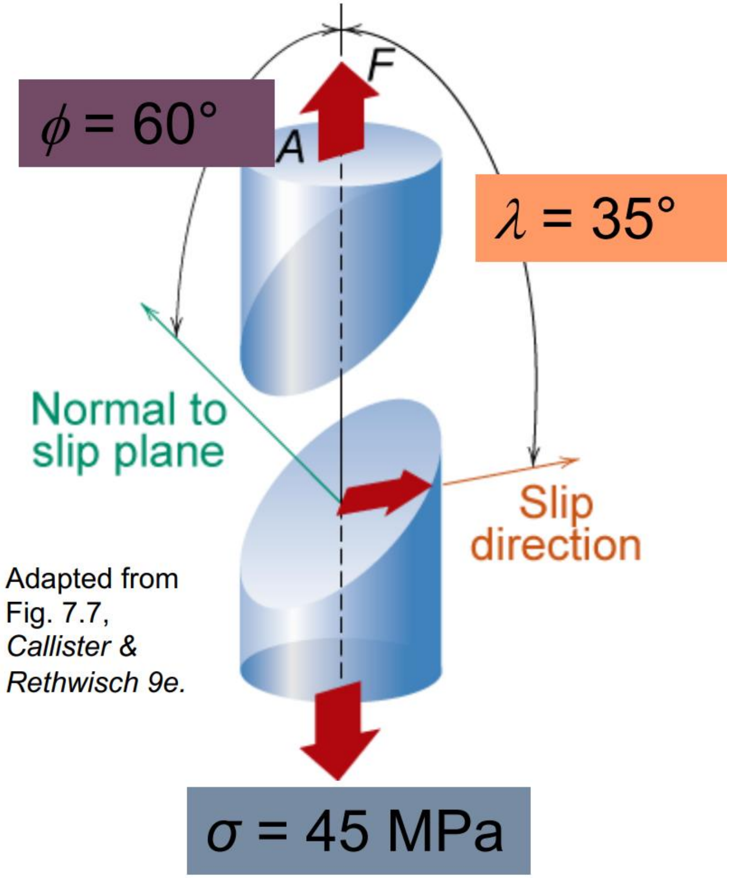
$$\boxed{\tau_R = \sigma cos\lambda \cos\phi = 45\cos(35°)\cos(45°) = 18.4\ MPa}$$

Since $\tau_R < \tau_{CRSS}$, no yield will occur.

b) For yielding to occur, $\tau_R \geq \tau_{CRSS}$. So, the normal stress σ_Y causing yielding by slip on those specific planes and direction can be found from the relationship:

$$\boxed{\tau_{CRSS} = \sigma_Y cos\lambda \cos\phi}$$

Substituting the values, one can obtain that yield can be induced for $\sigma \geq \sigma_Y = 50.5\ MPa.$



Adapted from Fig. 7.7, *Callister & Rethwisch 9e.*

Slip in single crystals

Example 2:

Consider a metal single crystal oriented such that the normal to the slip plane and the slip direction are at angles of 60° and 35°, respectively, with the tensile axis. If the critical resolved shear stress is 6.2 MPa (900 psi) determine:

a) if an applied stress of 12 MPa (1750 psi) cause the single crystal to yield;

b) If not, estimate the stress that would be necessary.

Hint: consider using equation for resolved shear τ_r.

Slip in single crystals

Example 2 solution:

Consider a metal single crystal oriented such that the normal to the slip plane and the slip direction are at angles of 60° and 35°, respectively, with the tensile axis. If the critical resolved shear stress is 6.2 MPa (900 psi) determine:

a) If an applied stress of 12 MPa (1750 psi) cause the single crystal to yield;

$$\boxed{\tau_R = \sigma cos\lambda \cos\phi = 12\cos(35°)\cos(60°) = 4.92\ MPa < 6.2\ MPa}$$

Yield will not occur.

b) If not, estimate the stress that would be necessary.

$$\boxed{\sigma = \frac{\tau_R}{cos\lambda \cos\phi} = \frac{6.2}{0.41} = 15.12\ MPa}$$

Slip in single crystals

Example 3:

A single crystal of zinc used for a tensile test is oriented such that the normal to its slip plane makes an angle of 65° with the tensile axis. Three possible slip directions make angles of 30°, 48°, and 78° with the same tensile axis.

(a) Which of these three directions slip is most favoured?

(b) If plastic deformation begins at a tensile stress of 2.5 MPa (355 psi), determine the critical resolved shear stress for zinc.

Hint: think about what would be the maximum angle where the slip will occur

Slip in single crystals

Example 3 solution:

A single crystal of zinc used for a tensile test is oriented such that the normal to its slip plane makes an angle of 65° with the tensile axis. Three possible slip directions make angles of 30°, 48°, and 78° with the same tensile axis.

(a) Which of these three directions slip is most favoured?

Slip will occur along the direction for which $cos\lambda\cos\phi$ is maximum:

cos(30°) = 0.87
cos(48°) = 0.67
cos(78°) = 0.21

Therefore, slip will occur along the direction $\lambda = 30°$.

b) If plastic deformation begins at a tensile stress of 2.5 MPa (355 psi), determine the critical resolved shear stress for zinc.

$$\boxed{\tau_{CRSS} = \sigma_Y cos\lambda\cos\phi = 2.5\cos 30\cos 65 = 0.91\text{MPa}}$$

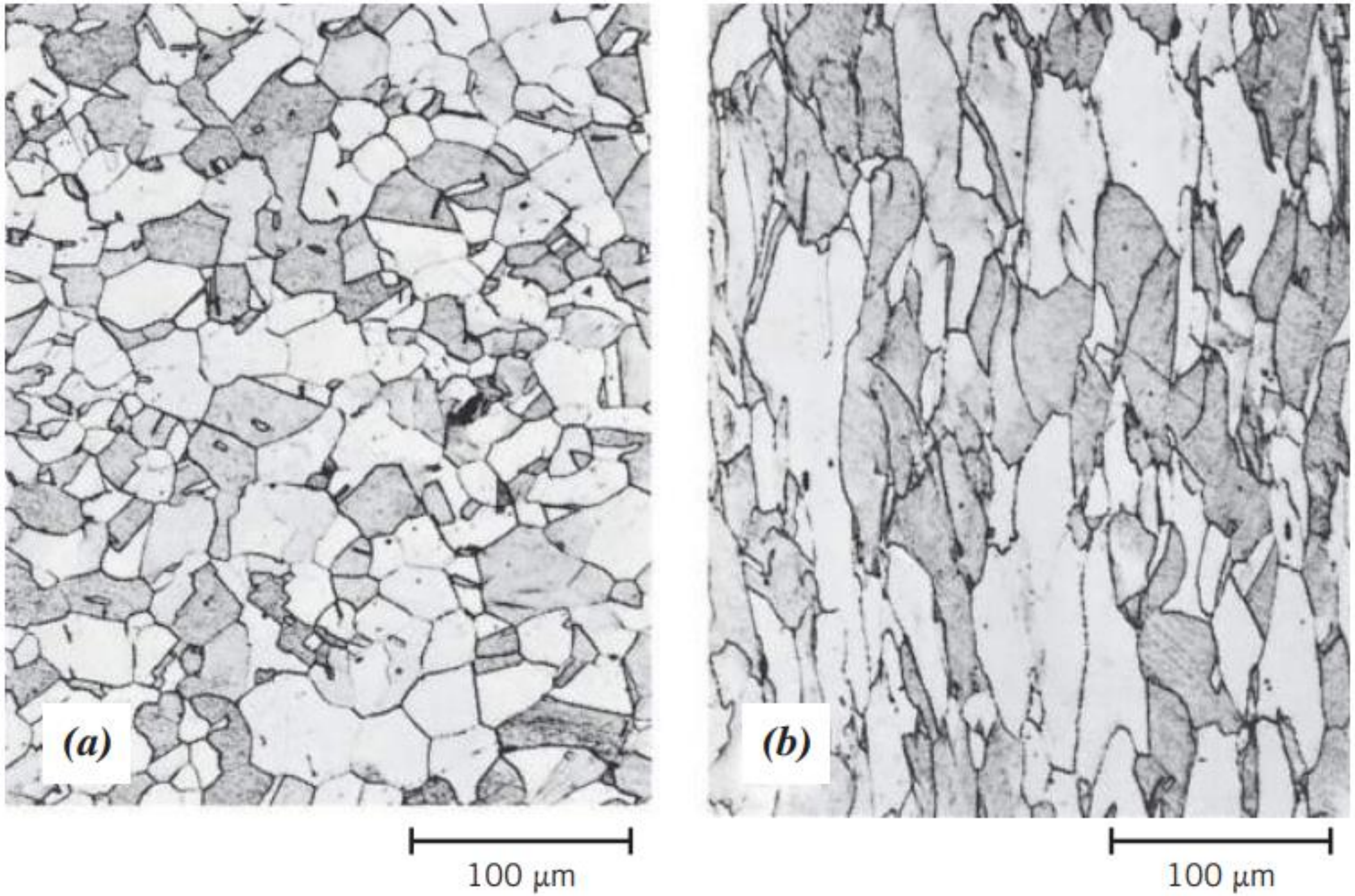
Slip in polycrystals

- Deformation and slip in polycrystalline materials is more complex than in single crystals.
- Because of the random crystallographic orientations of the numerous grains, the direction of slip planes and slip directions vary from grain to grain.
- The critical resolved shear stress also varies in different grains.
- The grains in which the resolved shear stress will be highest will yield first.
- Other less-favourably oriented grains will yield later.

Adapted from Fig.7.10, Callister & Rethwisch 9e.

Slip in polycrystals

Mechanical integrity is maintained during deformation, and each grain is constrained by its surrounding grains.

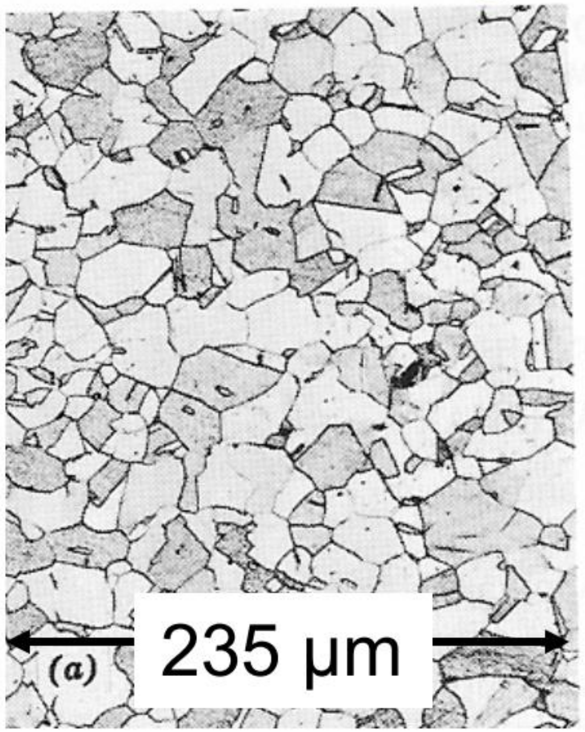


Under a tensile stress of sufficient magnitude, grain morphology may change from equiaxed (before deformation) to elongated along the stress axis (during plastic deformations).

Anisotropy

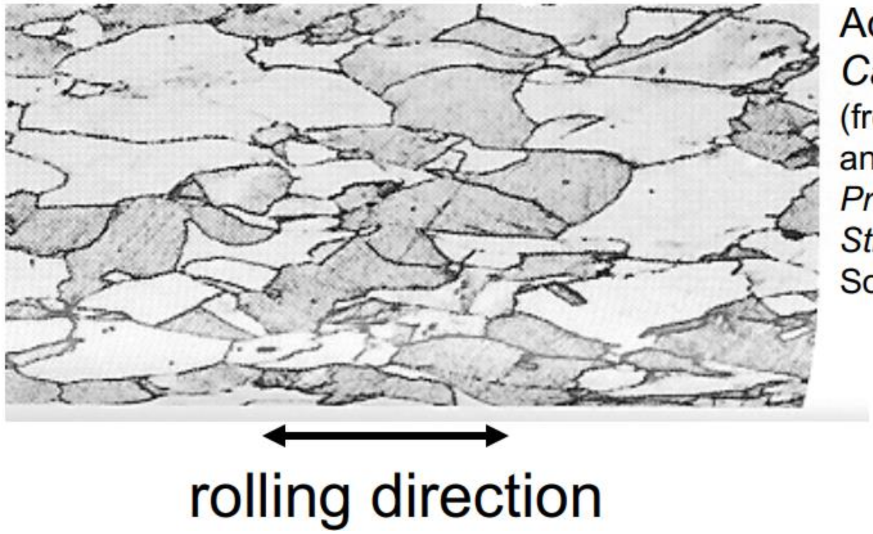
Anisotropy in mechanical properties, but also in other properties, can be induced by rolling, which induces a microstructure with elongated grains.

Before rolling



The mechanical response is isotropic, since grains are equiaxed and randomly oriented.

After rolling



Adapted from Fig. 7.11, *Callister & Rethwisch 9e.* (from W.G. Moffatt, G.W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, p. 140, John Wiley and Sons, New York, 1964.)

The mechanical response after rolling would be anisotropic, since grains are oriented along the rolling direction.

How can we make metallic materials stronger?

Strengthening strategies

Materials scientists and engineers are often required to design materials with high strengths, whilst being ductile and tough. Usually, ductility decreases when strength increases.

Because hardness (resistance to plastic deformations) and strength (both yield and tensile) are based on how easy plastic deformation occurs, by reducing the mobility of dislocations, the mechanical strength may be enhanced. By hindering dislocation motion, greater mechanical forces will be required to initiate plastic deformation.

Strengthening techniques mainly rely on the general principle, according to which restricting or hindering dislocation motion renders a material harder and stronger.

Strengthening strategies

How can we make metallic materials more resistant to plastic deformations?

A) Reducing grain size;
B) By additions of appropriate alloying elements (solid-solution strengthening).

Effect of grain size on dislocation motion and yield stress

Grain boundaries are barriers to slip. The barrier strengths depends on the orientation of the neighbouring grains and the grain size:

- The strength of the grain boundary barriers increases with increasing the misorientation angle between grains:

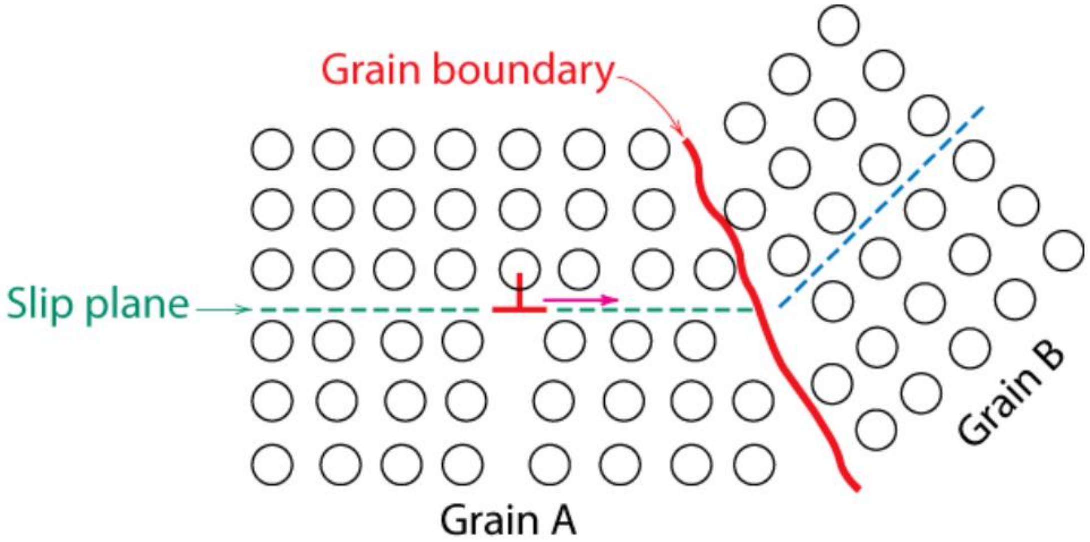


Fig. 7.14, *Callister & Rethwisch 9e.*
(From L. H. Van Vlack, *A Textbook of Materials Technology*, Addison-Wesley Publishing Co., 1973. Reproduced with the permission of the Estate of Lawrence H. Van Vlack.)

- Because of the atomic disorder within a grain boundary region, there will be discontinuities of slip planes from one grain into the other.

- A dislocation moving from grain A to grain B will have to change its direction of motion. This becomes more difficult as the crystallographic misorientation increases.

- Dislocations tend to "pile up" at grain boundaries. These pile-ups introduce stress concentrations ahead of their slip planes, which generate new dislocations in adjacent grains.

Effect of grain size on yield stress: Hall-Petch relationship

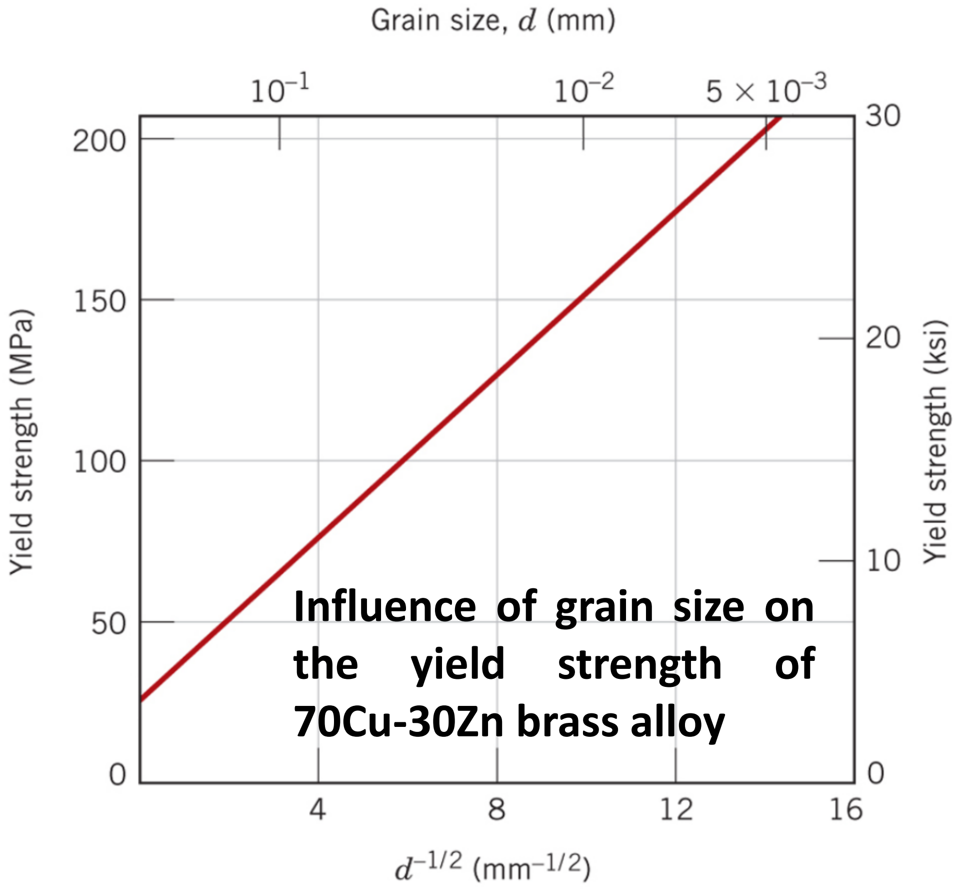
- Reducing the grain size, the amount of grain boundaries increases; hence, the hindrance to the slip process will increase.

- Smaller grain size leads to an increased yield stress, because dislocation movement become more difficult.

- The dependence of the yield stress on grain size is expressed by the **Hall-Petch relationship**:

$$\sigma_{yield} = \sigma_0 + k_y d^{-1/2}$$

d = average grain diameter
σ_0 = intercept
k_y = constant



Influence of grain size on the yield strength of 70Cu-30Zn brass alloy

Effect of grain size on yield stress

Example 4:

The lower yield point for an iron that has an average grain diameter of 1×10^{-2} mm is 230 MPa (33,000 psi). At a grain diameter of 6×10^{-3} mm, the yield point increases to 275 MPa (40,000 psi). At what grain diameter will the lower yield point be 310 MPa (45,000 psi)?

Hint: Look at all the data you have and compare it to find a solving equation

Effect of grain size on yield stress

Example 4 solution:

σ_y	d (mm)	$d^{-1/2}$ (mm)$^{-1/2}$
230 MPa	1×10^{-2}	10.0
275 MPa	6×10^{-3}	12.91

The given data allows to establish a system of 2 equations with 2 unknowns

$$230 \text{ MPa} = \sigma_0 + (10.0)\, k_y$$

$$275 \text{ MPa} = \sigma_0 + (12.91)\, k_y$$

Solving these two simultaneous equations results on

$\sigma_0 = 75.4$ MPa

$k_y = 15.46$ MPa(mm)$^{1/2}$

Substituting this values into the Hall-Petch relationship, we can calculate the grain size that would lead to a yied point of 310 MPa:

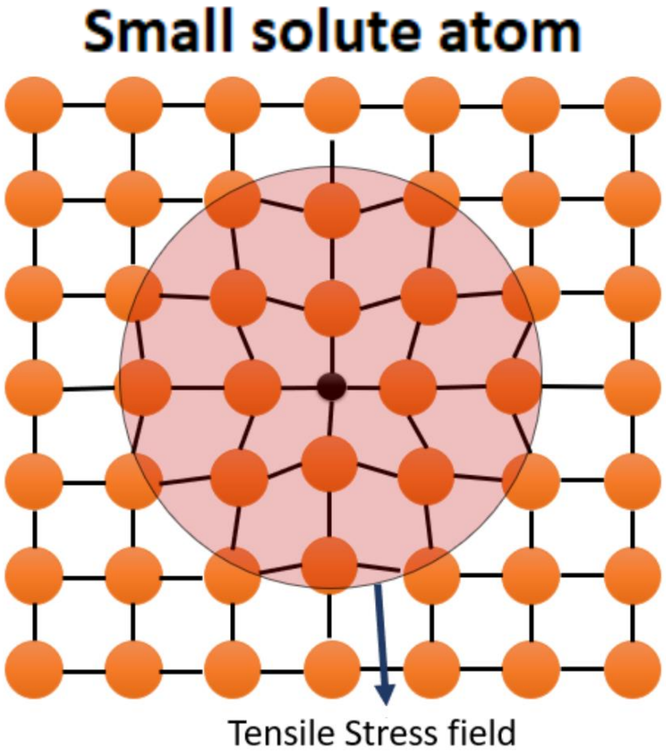
$$310 = 75.4 + \frac{15.46}{\sqrt{d}}$$

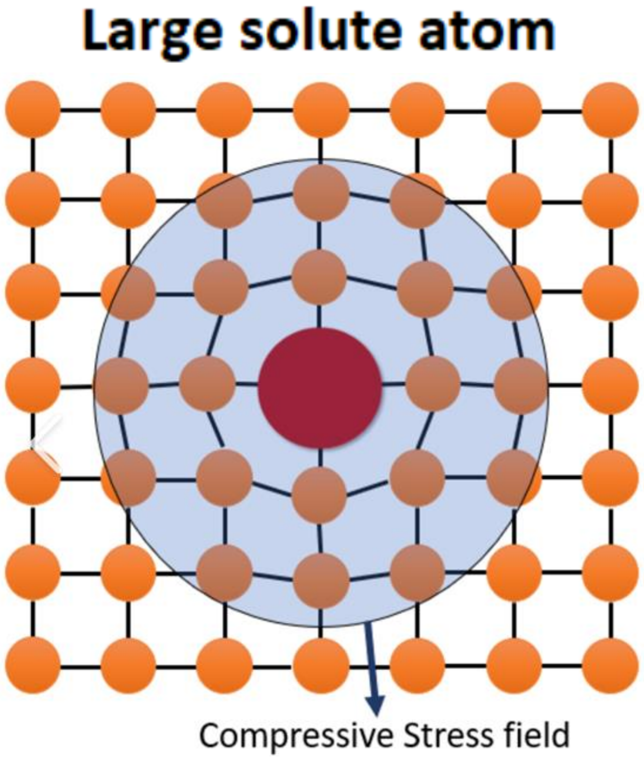
d = 0.00434 mm = 4.34 x 10^{-3} mm

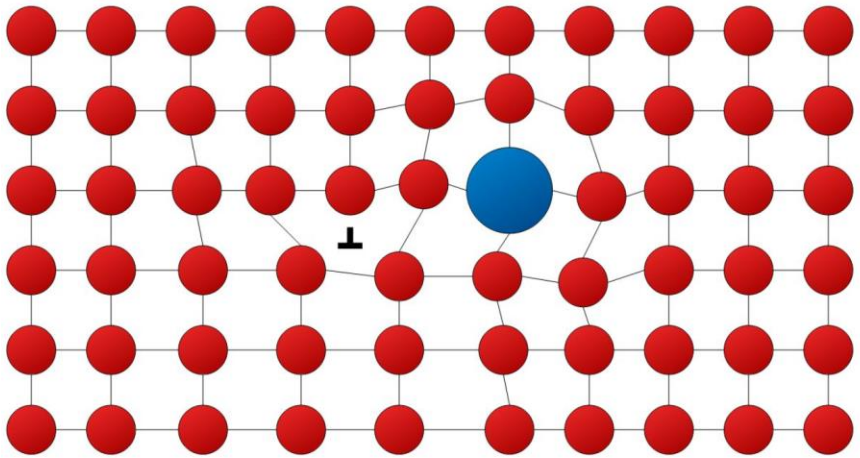
Solid-solution strengthening

- Hardening/strengthening can be also achieved by adding alloying elements or impurities to a pure metal.

- These elements can occupy regular lattice positions, vacancies, or interstitials, and depending on their atomic radii, they can induce a stress state in the lattice, that could result in a strengthening effect.



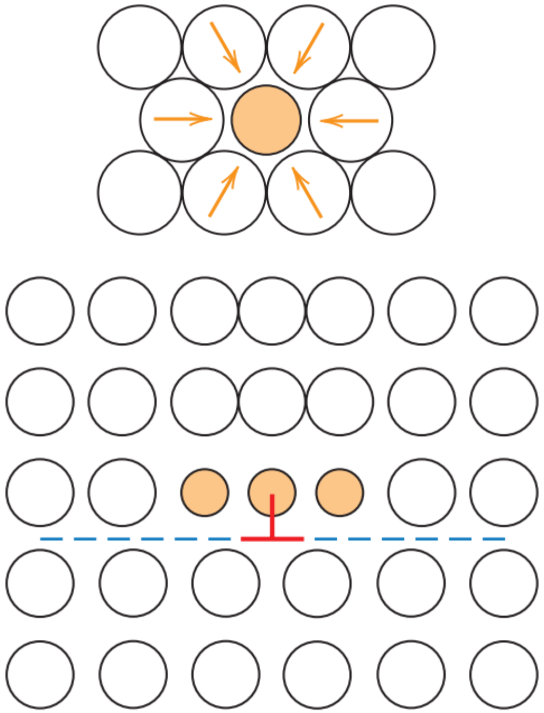


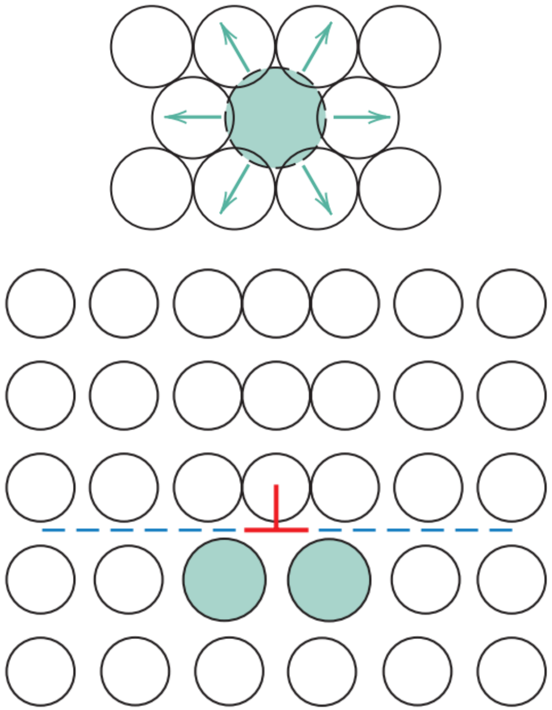


Solid-solution strengthening

Solute atoms tend to locate in lattice positions in a way to alleviate the existing stress state; for example:

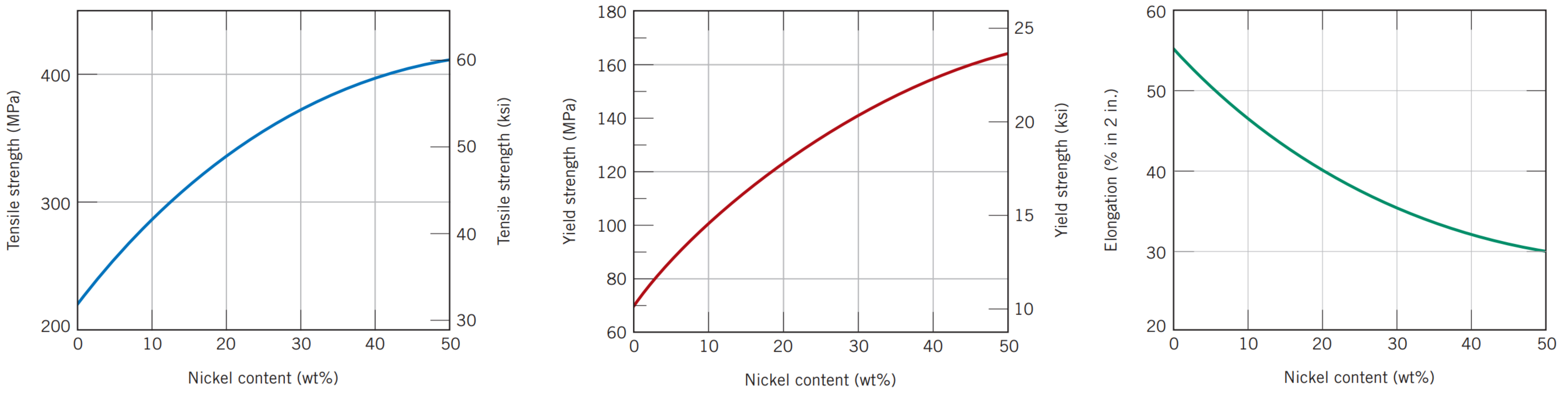
- Smaller solute atoms, which cause tensile stress in the surrounding local lattice, would locate in the compressive stress zone around a dislocation.
- Larger solute atoms, which cause compressive stress in the surrounding local lattice, would locate in the tensile stress zone around a dislocation.





Solid-solution strengthening

Effect of Nickel addition to copper metal.

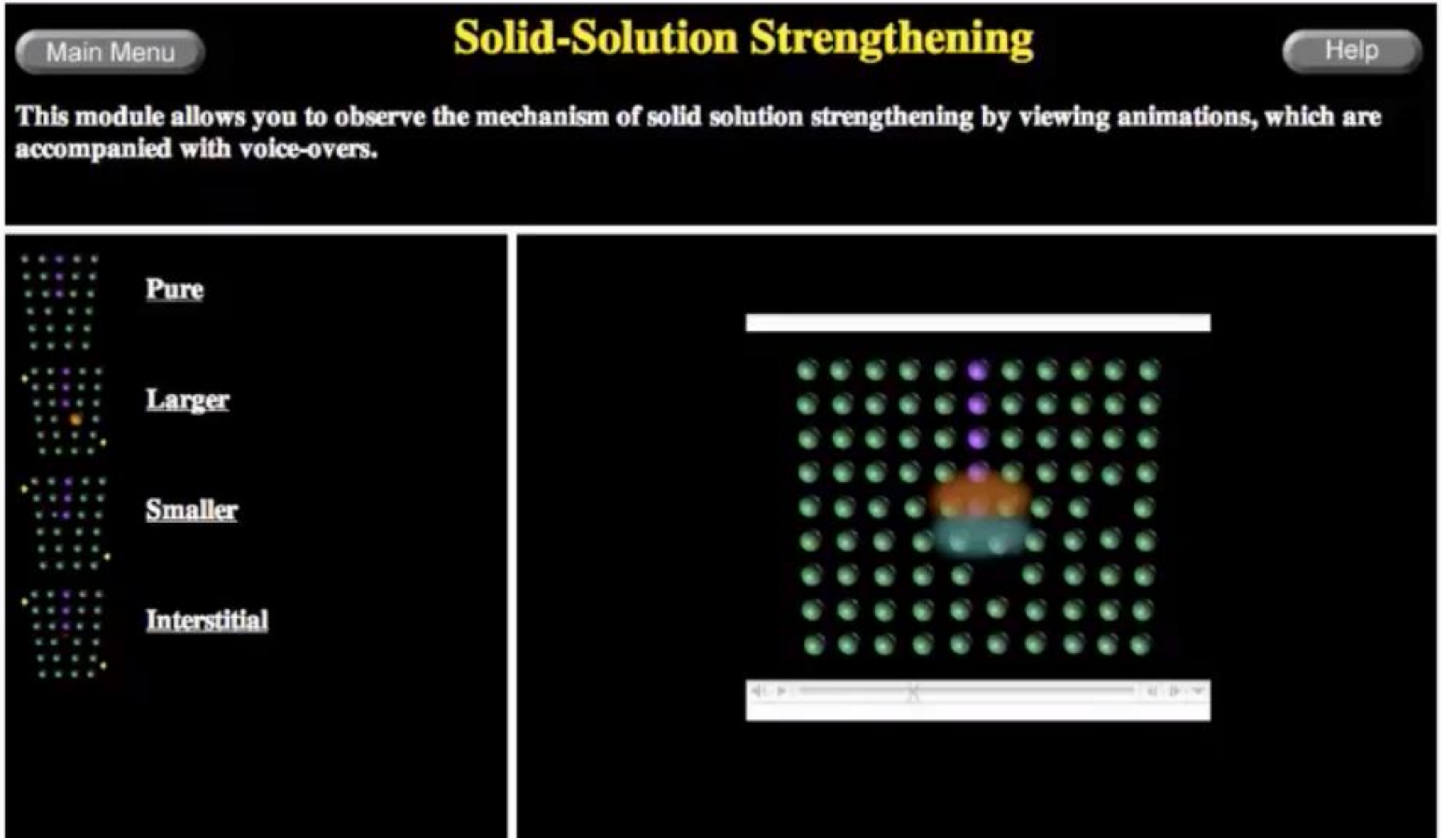


- Nickel additions increase tensile strength and yield point. The increase can be described by a power law ($\sigma{\sim}C^{1/2}$, where C is Ni content).

- Meanwhile, Ni additions reduces ductility of Cu.

Solid-solution strengthening

- Selection of solute atoms and their strengthening effects can be also evaluated using the VMSE environment.

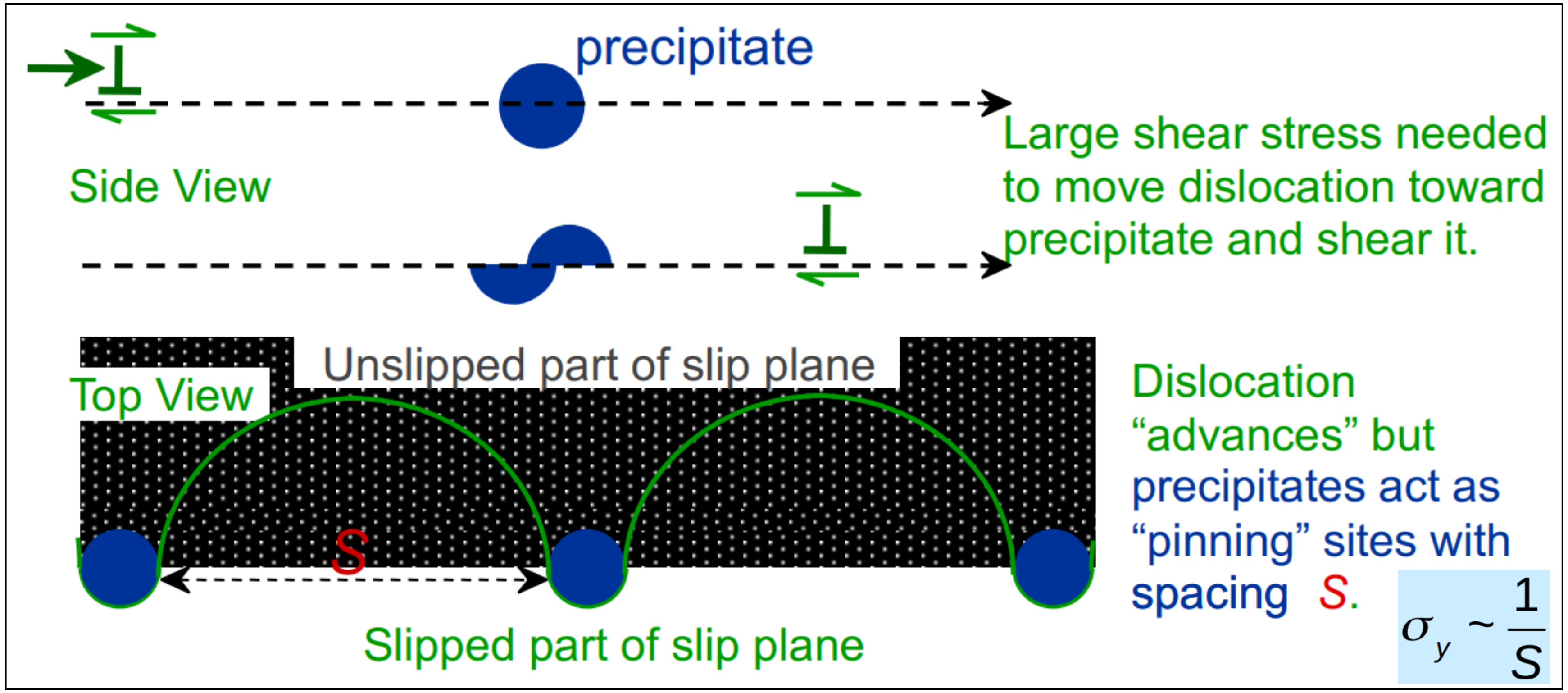


[Wiley - VMSE (drbuc2jl8158i.cloudfront.net)](https://drbuc2jl8158i.cloudfront.net)

Precipitation strengthening/hardening

Precipitation hardening, often called age hardening or particle hardening, is a heat treatment technique that drives the formation of small, uniformly dispersed particles (precipitates) of a second phase within the original phase matrix, as a result of a phase transformation, chemical reactions, etc.

The second phase particles should present increased obstacles for dislocation movement, to enhance the strength and hardness of metal alloys.

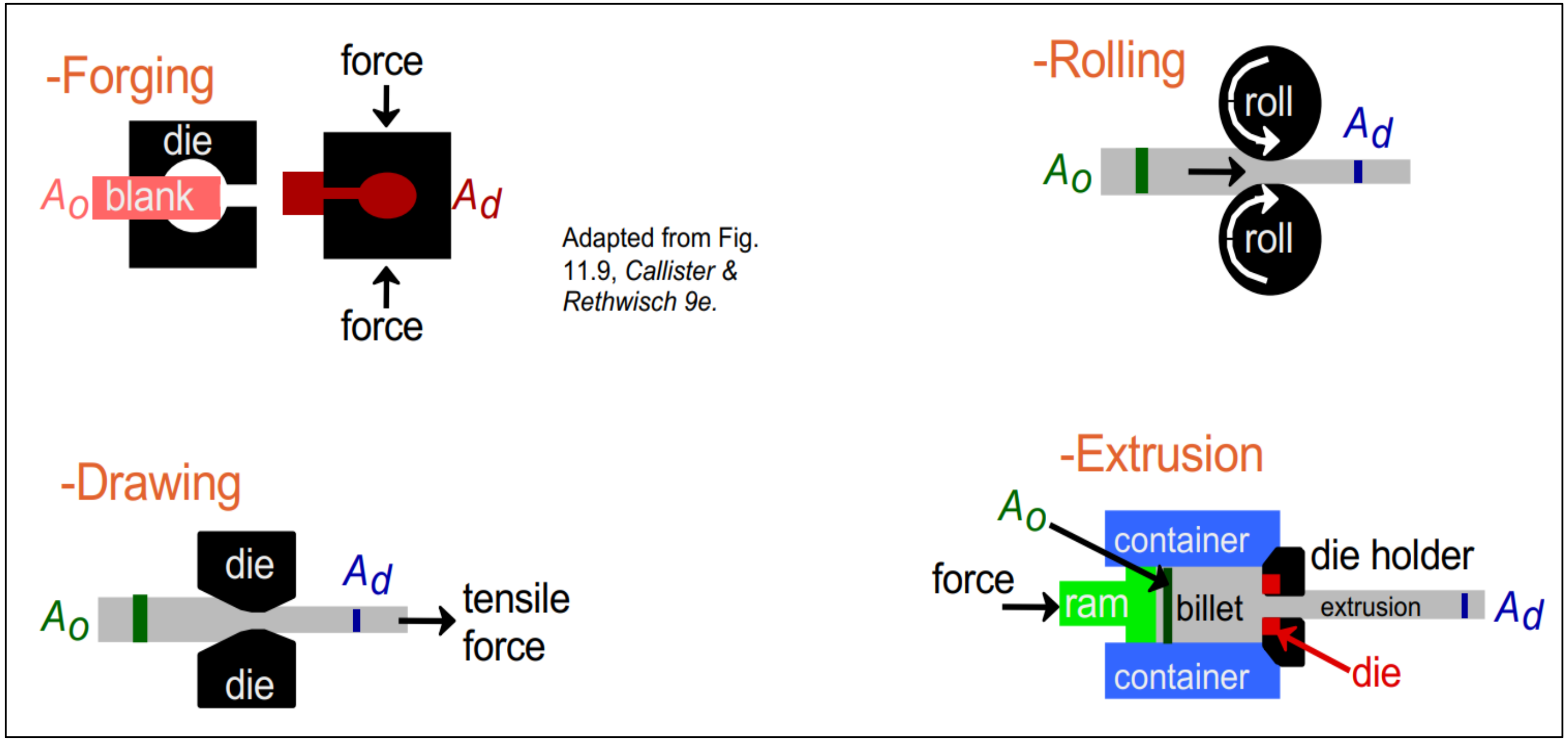


Examples of alloys that are hardened by precipitation treatments include aluminum–copper, copper–beryllium, copper–tin, and magnesium–aluminium. Other alloys based on nickel, titanium, some steels and ferrous alloys are also precipitation hardenable.

Cold working

Cold working produces an increase in strength but a decrease in ductility, because the metal strain hardens. There are 4 main cold work processes that can be carried out:

A metal piece is mechanically forced to assume a certain shape by mechanical forces.



Rolling, the most widely used deformation process, consists of passing a piece of metal between two rolls.

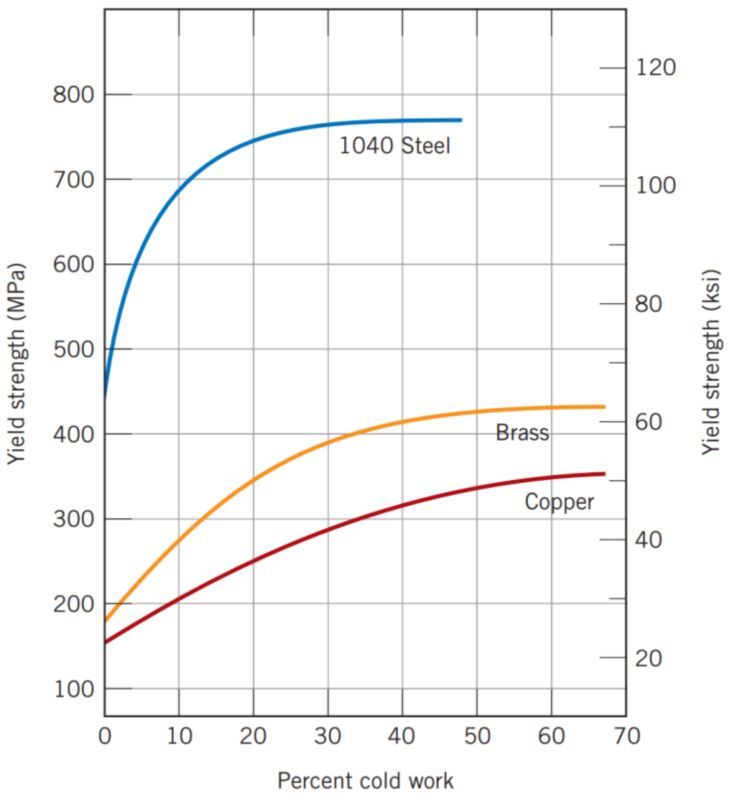
A metal piece is pulled through a die having a tapered bore by a tensile force. A reduction in cross section results, with a corresponding increase in length. Rod, wire, and tubing products are commonly fabricated in this way.

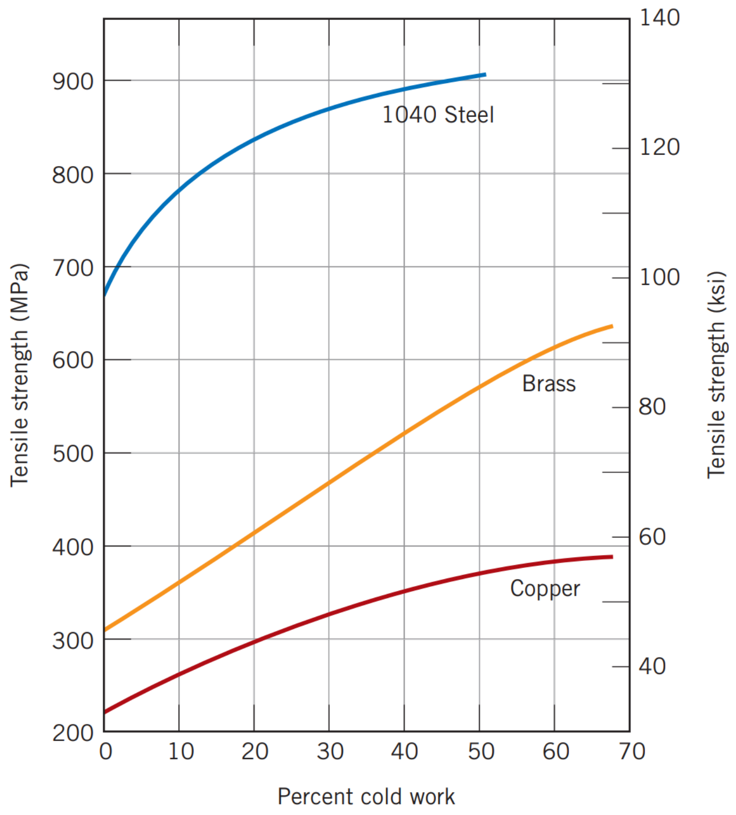
$$\%CW = \frac{A_o - A_d}{A_o} \times 100$$

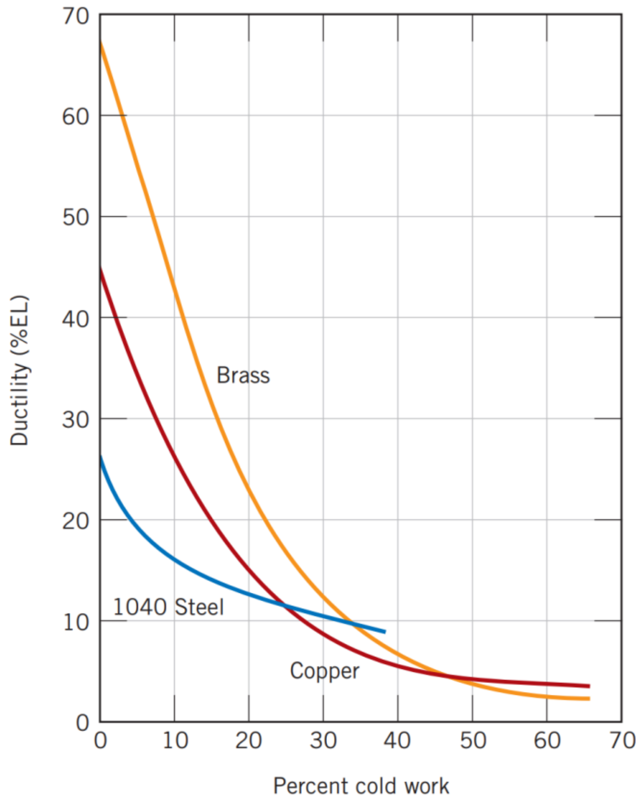
A bar of metal is forced through a die orifice by a compressive force that is applied to a ram.

Cold working

Effect of cold working on mechanical properties in different materials:

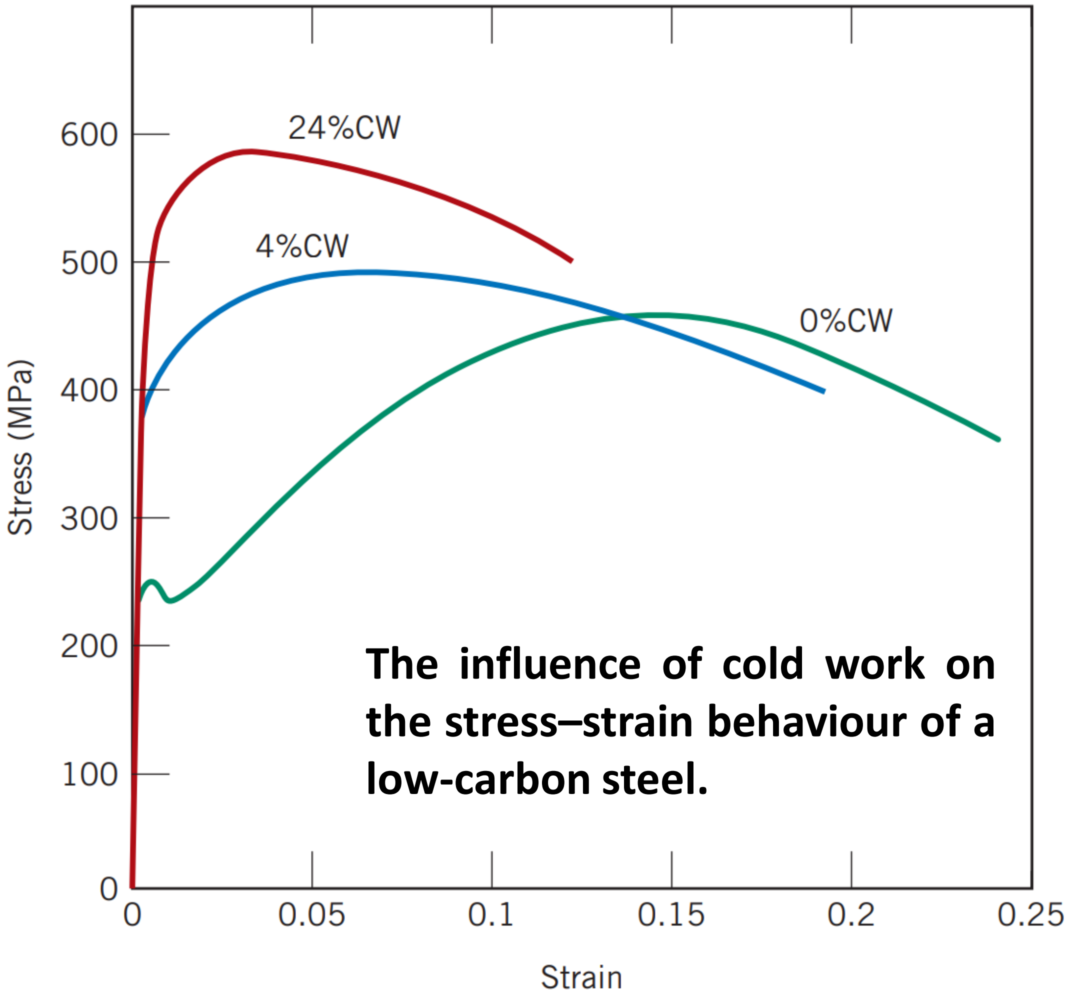






Cold working

Effect of cold working on mechanical properties in different materials:



The influence of cold work on the stress–strain behaviour of a low-carbon steel.

The phenomenon by which a ductile metal becomes harder and stronger as it is plastically deformed is also referred to as **strain hardening.**

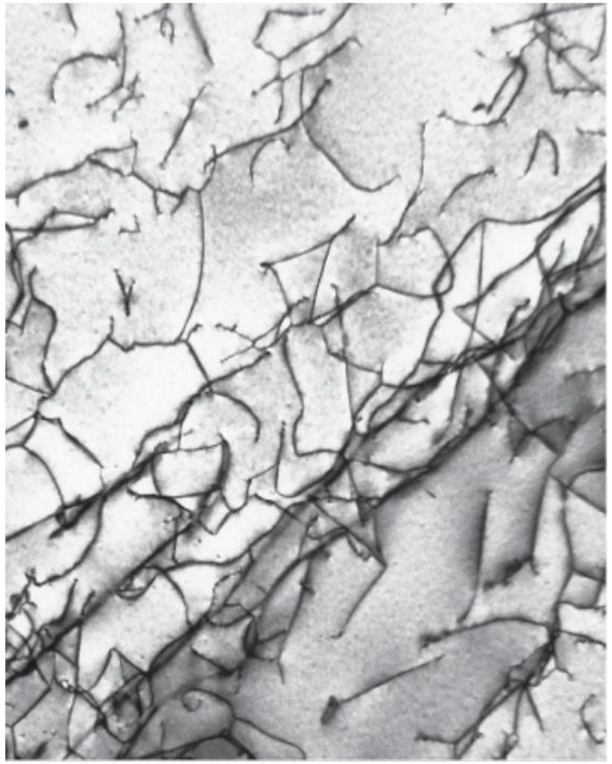
The dislocation density in a metal increases with deformation or cold work, due to the multiplication or formation of new dislocations.

On average, dislocation–dislocation strain interactions are repulsive. The net result is that the motion of a dislocation is hindered by the presence of other dislocations.

As the dislocation density increases, this resistance to dislocation motion by other dislocations increases.

Effect of cold working

Dislocation Structures Change During Cold Working



0.2 μm

Dislocation structure in Ti after cold working.

- Dislocations entangle with one another during cold work.
- Dislocation motion becomes more difficult.

Fig. 4.7, *Callister & Rethwisch 9e.* (Courtesy of M.R. Plichta, Michigan Technological University.)

Dislocation Density Increases During Cold Working

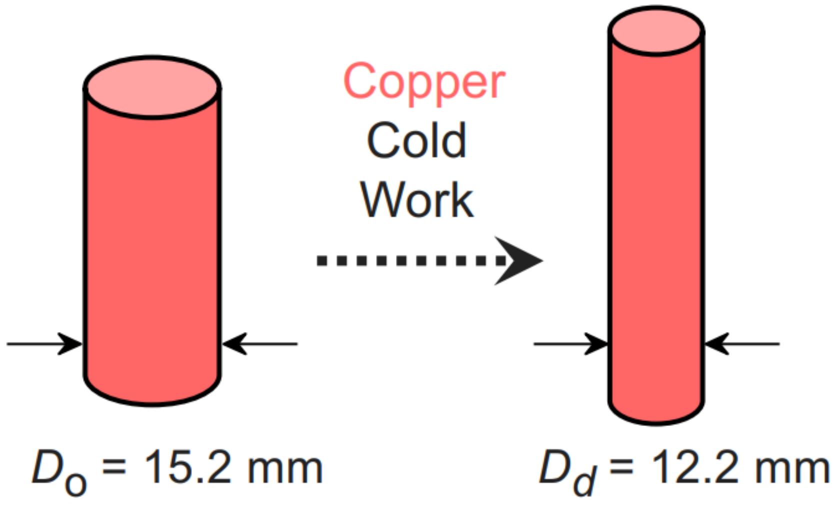
$$\text{Dislocation density} = \frac{\text{total dislocation length}}{\text{unit volume}}$$

- Carefully grown single crystals
  → ca. 10^3 mm^{-2}
- Deforming sample increases density
  → 10^9-10^{10} mm^{-2}
- Heat treatment reduces density
  → 10^5-10^6 mm^{-2}

Cold working

Example of calculation of cold working in copper:

What are the values of yield strength, tensile strength and ductility after cold working Cu?



$$\%CW = \frac{\dfrac{\pi D_o^2}{4} - \dfrac{\pi D_d^2}{4}}{\dfrac{\pi D_o^2}{4}} \times 100$$

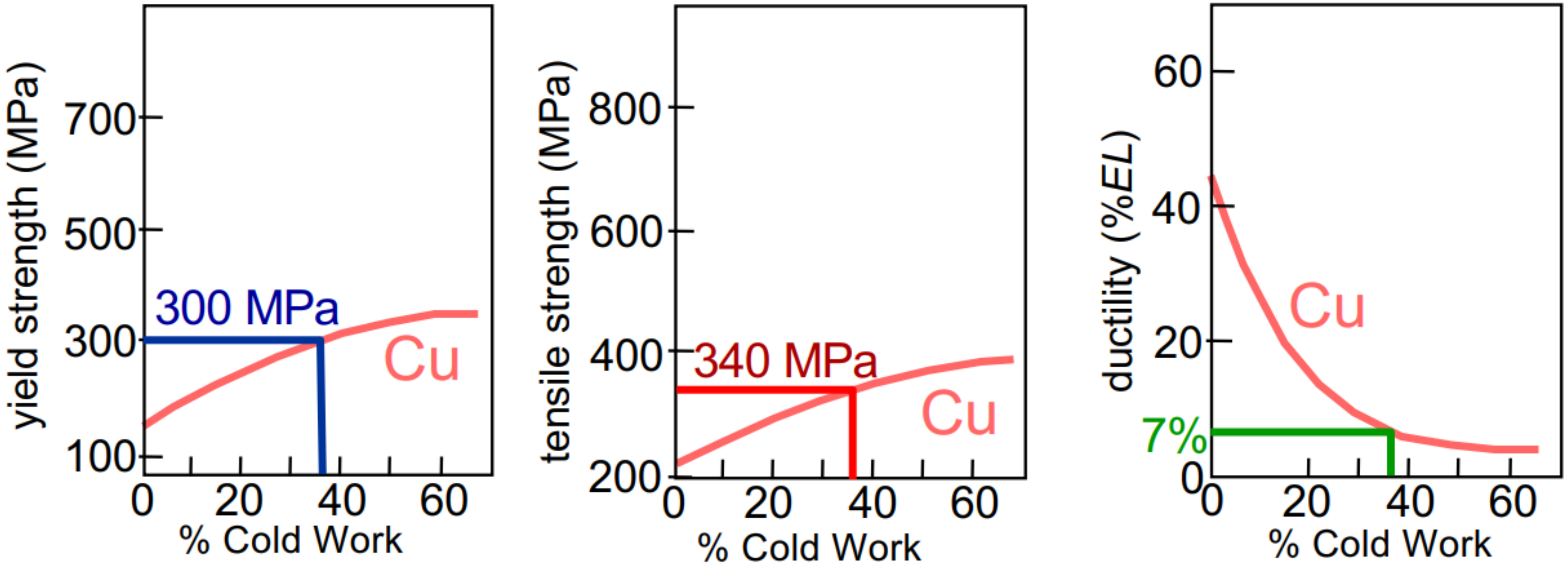
$$= \frac{D_o^2 - D_d^2}{D_o^2} \times 100$$

$$\%CW = \frac{(15.2\ \text{mm})^2 - (12.2\ \text{mm})^2}{(15.2\ \text{mm})^2} \times 100 = 35.6\%$$

Cold working

Example of calculation of cold working in copper:

What are the values of yield strength, tensile strength and ductility after cold working Cu?



σ_y = 300 MPa

TS = 340 MPa

%*EL* = 7%

Summary

- Dislocations are observed primarily in metals and alloys.

- Strength is increased by making dislocation motion difficult.

- Strength of metals may be increased by:

- decreasing grain size
- solid-solution strengthening
- precipitation hardening
- cold working